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Quantitative Investment Strategies

Quantifying Asymmetry –
Realized Skewness as a Core
Portfolio Strategy

In this article we present our new cross-asset Realized Skewness strategy. The strategy shows strong returns in backtesting and is a good natural complement to Trend Following.

We introduce a cross-asset Realized Skewness strategy designed to systematically capture risk premia across equities, rates, FX, and commodities:

The strategy leverages both persistent risk premia, associated with assets exhibiting long-term positive/negative skewness, and mean-reversion effects that typically follow large drawdowns. By constructing a long/short portfolio and scaling positions according to realized skewness signals, the approach maintains equal risk across asset classes and adapts dynamically to changing market conditions.

Performance analysis demonstrates strong risk-adjusted returns: During our backtest period starting from mid-2012, the strategy achieves a Sharpe ratio of approximately 1.01, and limited drawdown risk. The strategy is particularly robust in equities, rates, and commodities, while FX results are somewhat weaker but provide valuable diversification, especially during periods of market stress.

A key advantage of the Realized Skewness strategy is its low correlation with traditional Trend Following strategies: This makes it an effective diversifier within multi-strategy portfolios. When combined with Trend Following, the overall portfolio Sharpe ratio improves further to 1.15, and resilience during market crises is enhanced, as the two strategies tend to offset each other's weaknesses in different market regimes.

Extensive robustness tests confirm the stability of the strategy: We experiment with various signal definitions, market beta constraints, and weighting methodologies, ensuring that each choice is carefully considered to best capture the intended risk premia. Across these tests, the strategy consistently maintains strong risk-adjusted performance under a wide range of parameterizations, demonstrating resilience to different implementation choices and market environments.

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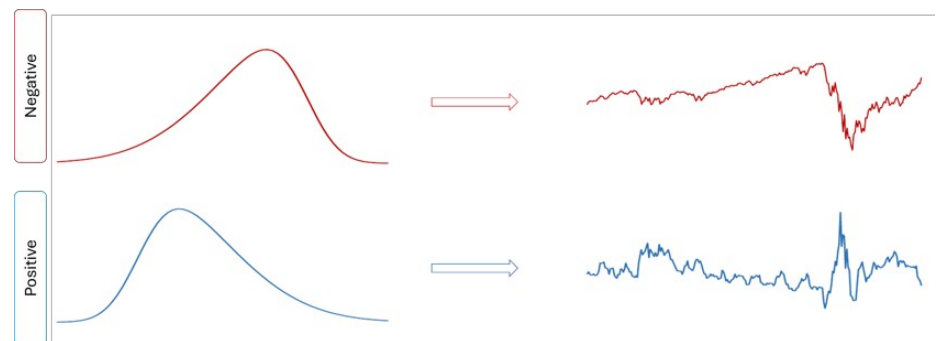
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Introduction

In this publication, we introduce our novel Realized Skewness strategy. The strategy is driven by the risk premia embedded in assets which exhibit negative skewness and benefits further from a mean-reversal effect after large drawdowns. We implement the skewness concept across asset classes and find a diversification effect from a cross-asset class implementation. The strategy is implemented in a long/short context, allowing it to take time-varying directional positions. This feature makes the strategy an interesting complement to cross-asset Trend Following, which shares design features in portfolio construction and signal design but delivers a different performance profile. We find our results to be robust in a range of stability tests.

Return skewness is a statistical measure calculated on realized returns which gives us an indication of the propensity of an asset to have large positive or negative returns where the sign of the skewness metric indicates the prevalent direction of such large returns. Investors have an aversion to large negative returns, which has been a topic of behavioral finance and utility function modeling (see, for example, the Prospect Theory by Kahneman and Tversky, 1979). Thus, we would expect a risk premium associated with assets where the skewness is negative and has a large magnitude. Similarly, skewness tends to be negative in times of market corrections, and relates to a mean-reversal effect in drawdowns. This latter effect is of a more tactical nature while the prior effect may be a more persistent premium attached to an asset. Both drivers essentially indicate that the expected return of an asset should increase as the skewness becomes smaller.

Exhibit 1: Illustration of price movements with negative and positive skewness

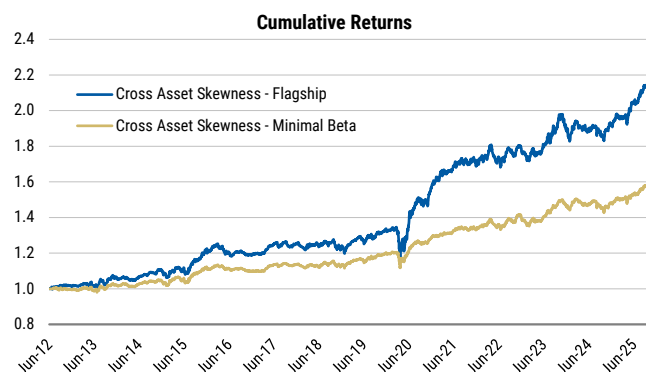


Source: Morgan Stanley Research

We take advantage of this effect by constructing a strategy that takes a range of assets across equities, rates, FX, and commodities and scale it by a modified skewness metric. The modification is quite simple: we multiply the realized empirical skewness by -1 and then cap/floor the resulting total score at +/-1. This modified metric then increases short positions in assets where the skewness is positive and increasing. Where the skewness is negative and decreasing we take larger short positions. We then impose additional constraints that portfolio weights are capped so that, before volatility targeting, beta exposure to each asset class remains within $\pm 30\%$, and turnover is actively managed to avoid excessive trading costs.

We find the resulting metric to deliver a strategy with strong risk-adjusted returns. The Sharpe ratio is 1.01 with a persistently positive return, particularly over the more recent past. While we seek long exposures to assets with negative skewness, the drawdown risk of the resulting strategy is actually limited: the maximum drawdown is just 2.37 annual standard deviations, which is in line with other long/short strategies. We also find limited exposures to traditional long-only assets. For the cumulative returns as well as key statistics of the strategy, see [Exhibit 2](#) and [Exhibit 3](#).

Exhibit 2: Cross-asset Skewness performance



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 3: Performance statistics

	Cross Asset Skewness - Flagship	Cross Asset Skewness - Minimal Beta
CAGR	5.91%	3.51%
Annualized Vol	5.88%	3.66%
CAGR / Vol	1.01	0.96
Max DD	-13.95%	-7.47%
Max DD / Vol	-2.37	-2.04
Calmar Ratio	0.42	0.47
Sortino Ratio	1.45	1.40
VaR (1%, Monthly)	-3.83%	-2.26%
CVaR (1%, Monthly)	-6.58%	-3.46%
Hit Rate (Monthly)	62.45%	60.81%
Skewness	-0.41	-0.03
Excess Kurtosis	15.89	7.98

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

We also present a market-neutral variant of the strategy, by constraining market beta exposures within each asset sleeve to be close to 0. This adjustment reduces directional risk while preserving the core signal dynamics. The minimal beta approach maintains much of the performance observed in the original specification, albeit removing the exposure to market-level timing. Despite the lower-beta constraints, the strategy remains highly competitive, posting a Sharpe ratio of 0.96 – only marginally lower than the less constrained version. This demonstrates that the skewness-based signal retains its efficacy even under tighter neutrality conditions, offering an attractive option for investors seeking alpha without significant market exposure.

Our analyses also find a low correlation of the strategy with Trend Following (correlation of close to 0 on average). Given that Trend Following as well as our Skewness strategy belong to the class of directional long/short strategies, this feature makes them good complements to each other. Investors with exposure to Trend Following only should consider the addition of the Skewness strategy for diversification.

In our analyses we conduct a large range of stability analyses and find the strategy to be rather robust. There are no pin risks in the parametrization. When looking at the asset classes, we find strong performance of the Skewness strategy in rates, equities, and commodities. For FX the performance is less strong. We attribute this observation to the fact that in FX there is always one currency long and another short, making it harder to define the return direction in which investors would be exposed to losses.

The role of skewness in asset pricing has attracted sustained academic interest, evolving from theoretical foundations to robust empirical applications across asset classes. The seminal work by Kraus and Litzenberger (1976) introduced skewness preference into asset valuation, extending the Capital Asset Pricing Model (CAPM) to include third moments.

Their three-moment CAPM posited that investors not only care about mean and variance but also about the asymmetry of returns, laying the groundwork for decades of research into skewness premia.

Building on this foundation, Harvey and Siddique (2000) introduced conditional skewness into equity asset pricing tests, showing that assets with more negative tail risk command higher expected returns. Their model has been validated and extended in subsequent studies, including a 25-year out-of-sample evaluation (Harvey and Siddique, 2023) and a broader reassessment of systematic skewness (Anghel et al, 2022). These works confirm that skewness is a priced risk factor and robust across time and specifications.

Skewness is also relevant beyond equities. Fernandez-Perez et al (2018) showed that skewness is priced in commodity futures, and Jiang et al (2019) found skewness to have forecasting power for currency markets, suggesting that skewness is a universal feature of risk pricing. These extensions broaden the applicability of skewness-based models across asset classes. We can confirm the existence of a skewness premium in equities, commodities, and rates with positive but weak evidence in FX.

One interesting aspect about the literature on skewness signals is that there appears to be a fundamental difference between ex ante (i.e., implied) and ex post (i.e., realized) skewness. For realized skewness, authors such as Harvey and Siddique (2000, 2023), Angel et al, (2022), Fernandez-Perez et al (2018) and others find a positive risk premium associated with negative skewness. The signal power of realized skewness is also associated with a mean-reversal effect. Researchers that focus on ex ante skewness, such as Conrad et al (2013), Langlois (2020), or Rehman and Vilkov (2012), frequently find positive returns associated with positive implied skewness, highlighting the return forecasting ability of option traders as the driver of the effect. Xing et al (2010) add to this evidence by finding that the volatility smirk in individual stock options predicts future equity returns, while DeMiguel et al (2013) showed that incorporating option-implied skewness and volatility improves portfolio performance. These findings underscore that skewness generally is a concept that has high forecasting power for returns but the exact nature of this forecasting power depends on how exactly skewness is calculated – chiefly if it is an ex post or ex ante metric. In this article we focus on **realized** skewness as a metric to forecast expected returns (i.e., we focus on the risk premium and mean-reversal characteristics associated with assets that have negative realized skewness).

A complementary strand of literature offers behavioral interpretations. Bali et al (2009) argue that investors treat certain stocks like lotteries, preferring assets with positively skewed payoffs despite lower expected returns. Brunnermeier et al (2007) provide a theoretical model in which investors form optimal beliefs that favor skewed returns, offering a behavioral foundation for skewness premia. Bauer and Chernov (2024) extend this idea to interest rate markets, linking skewness to biased beliefs about monetary policy.

Finally, methodological contributions such as Kim and White (2004) have improved the robustness of skewness estimation. Our contribution in this space is to show that standard empirical skewness metrics achieve stronger performance than robust ones as robust metrics tend to dampen the measurement of the tails of the distribution which has high signaling power. Amaya et al (2015) demonstrated that realized skewness, derived from high-frequency data, predicts the cross-section of equity returns. Meta-analyses like

Jensen et al (2022) have raised concerns about replication in finance, prompting renewed scrutiny of skewness-based factors. That said, Jensen et al (2022) confirm a positive alpha for realized skewness signals.

In summary, the literature converges on the conclusion that skewness is a meaningful and priced dimension of risk. Whether measured conditionally, realized, or implied, skewness consistently predicts returns across equities, commodities, and currencies. The definition of skewness is relevant as the signal appears to have different drivers when calculated using an ex ante (implied volatility) or ex ante (realized) angle. The integration of behavioral insights and methodological advances has deepened our understanding of why skewness matters, while recent replication efforts have strengthened confidence in its empirical relevance. This body of work affirms that skewness is not merely a statistical artifact but a fundamental component of asset pricing.

For our purposes, this means that realized skewness is a relevant factor that can be applied across asset classes to select assets and generate risk premia. While we agree with the literature that skewness is related to a risk premium, we think that it is also related to a mean-reversal effect, particularly when skewness becomes rather negative in drawdowns and assets revert subsequently.

The remainder of this publication is structured as follows. In [Signal Characteristics and Performance Drivers](#) we discuss the characteristics of the skewness signal and run a couple of analyses on its predictive power for returns. Then in [Portfolio Construction and Strategy Performance](#) we move on to portfolio construction and a deep dive into the return and exposure characteristics of the Skewness strategy. Given its characteristics as a directional long/short strategy, we discuss the interactions between Trend and Skewness in [Interaction Between Skewness and Trend Following](#). We follow up this analysis with a discussion of a range of [Robustness Tests](#) showing that the strategy works well across a range of specifications. We then provide the final [Conclusions](#).

Signal Characteristics and Performance Drivers

In this section we focus on the definition of our skewness metric and its characteristics. We rely in our work on realized skewness, which is based on the realized returns of the underlying asset. Our definition follows the baseline sample skewness metric:

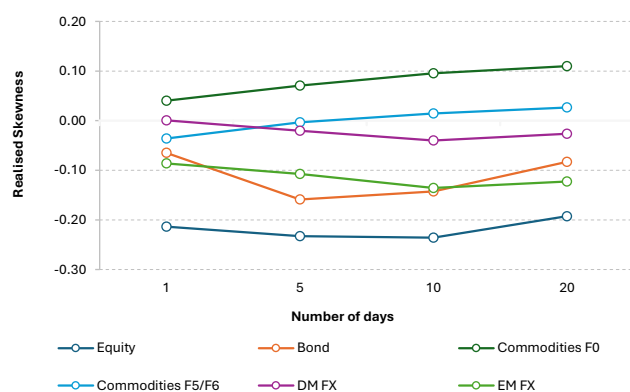
$$Skew_t^i = \frac{\sqrt{n(n-1)}}{n-2} \times \frac{\frac{1}{n} \times \sum_{k=0}^{n-1} (r_{t-k}^i - \mu_t^i)^3}{\left[\frac{1}{n} \times \sum_{k=0}^{n-1} (r_{t-k}^i - \mu_t^i)^2 \right]^{3/2}}$$

where r stands for the returns, μ stands for the expected return and n for the estimation horizon we consider. We calculate this skewness metric over different estimation windows and for different return periods to get more stability.

We show the average skewness by asset class and for different return periods ([Exhibit 4](#)) as well as different estimation windows ([Exhibit 5](#)). Thus, the skewness numbers are averaged across assets within an asset class as well as time. For all the cases the average ranges from -0.25 to +0.15, which shows that realized skewness is on average close to zero within a narrow range. Thus, it's not unreasonable to assume that empirical skewness is roughly anchored around zero.

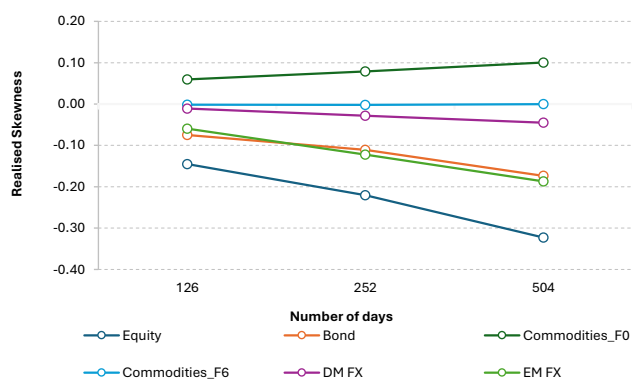
We also find that skewness of daily returns has a negative bias, particularly for equities, while we get to a balanced view of skewness around 20-30 days across the asset classes. Another feature of skewness is that it tends to diverge between asset classes depending on the estimation window we are looking at. Using longer estimation windows, such as 504 days, we see skewness in equities drop from an average of -0.15 to below -0.3.

Exhibit 4: Realized skewness by return windows



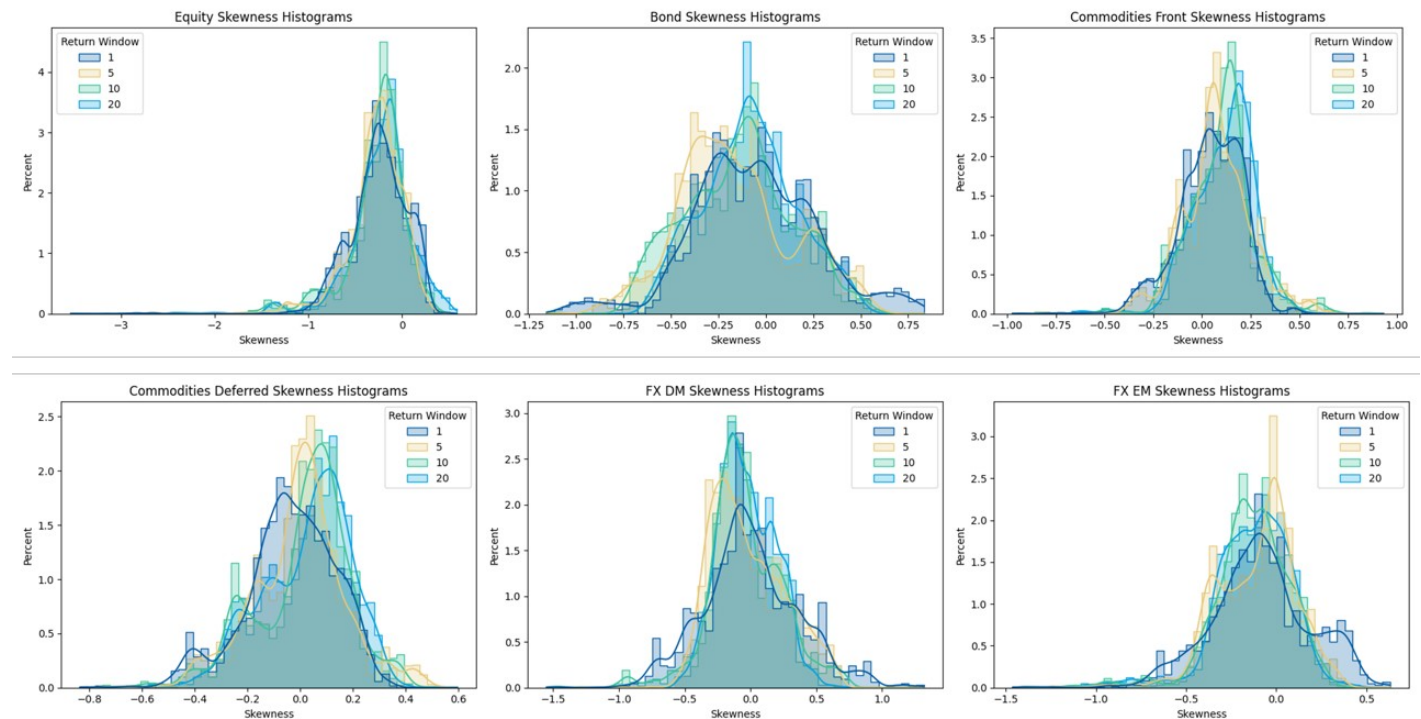
Source: Morgan Stanley Research

Exhibit 5: Realized skewness by lookback windows



Source: Morgan Stanley Research

The empirical distributions of the skewness metrics is illustrated in [Exhibit 6](#). We find skewness to generally range from -1.0 to +1.0. In the case of equities there is a left tail which goes out to below -3, indicating the tendency for equities to have crash risk. In terms of range we find the lowest dispersion of the skewness metrics for commodities.

Exhibit 6: Histograms of realized skewness

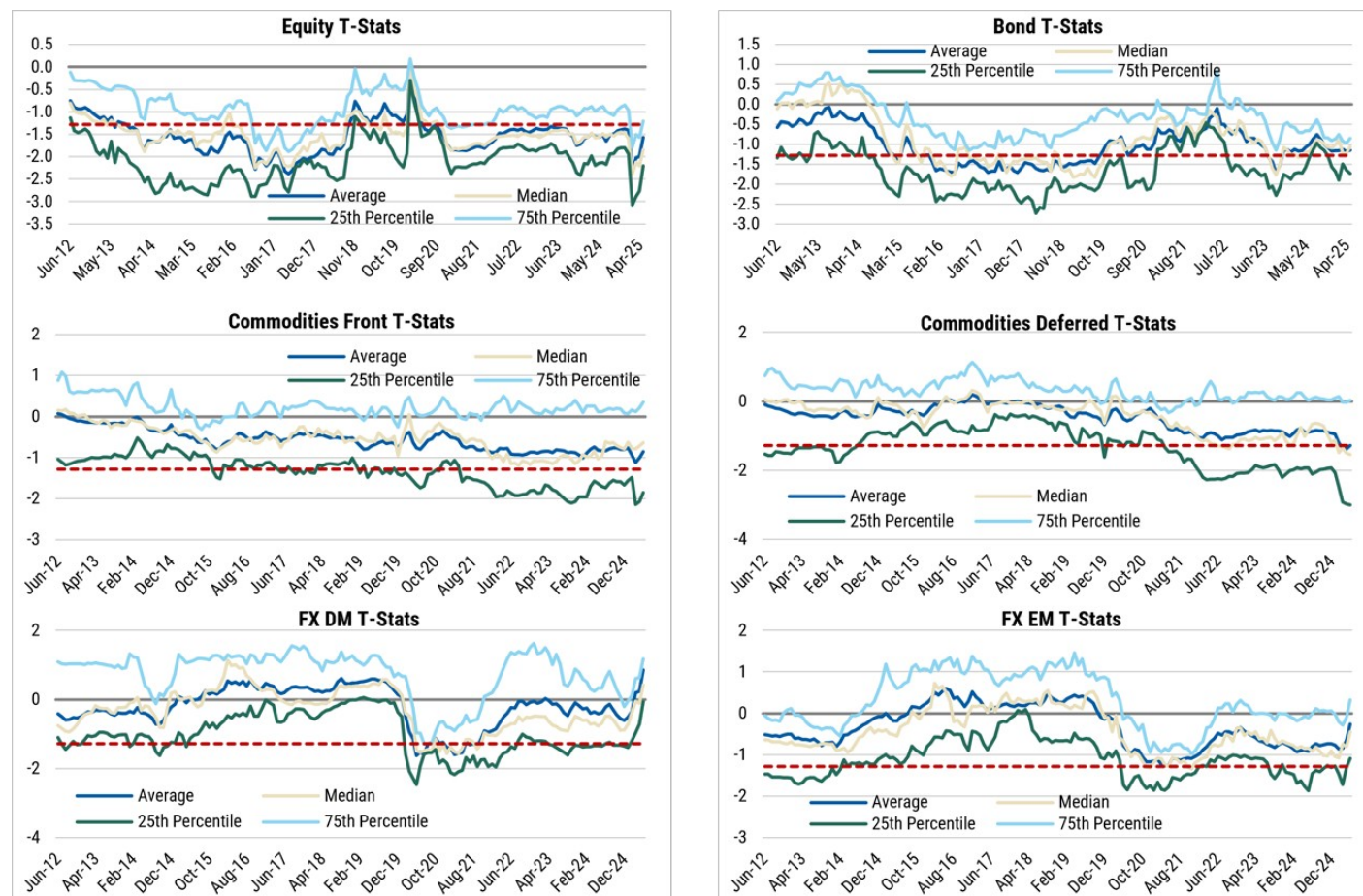
Source: Morgan Stanley Research

The considerations so far show that skewness is reasonably bound and centered close to zero. Between skewness and expected returns we find the negative link academics highlight in theoretical and empirical work (see the literature survey in [Introduction](#)). We plot in [Exhibit 7](#) the link between expected returns after scaling to 10% volatility and skewness scores. There is an apparent downward slope when we fit a line through the individual data points. The only asset class where the pattern is rather weak is in FX, where we would expect performance of a skewness strategy to be the most challenged based on these results. The return expectations difference between periods following the most negative and most positive skewness measurements goes up to 4% (the lower the realized skewness, the larger the average future returns), indicating that realized skewness and the following expected returns have an economically significant link. This pattern is essential to construct a profitable systematic trading strategy on the back of the realized skewness metric.

Exhibit 7: Realized skewness versus 1-month forward return

Source: Morgan Stanley Research

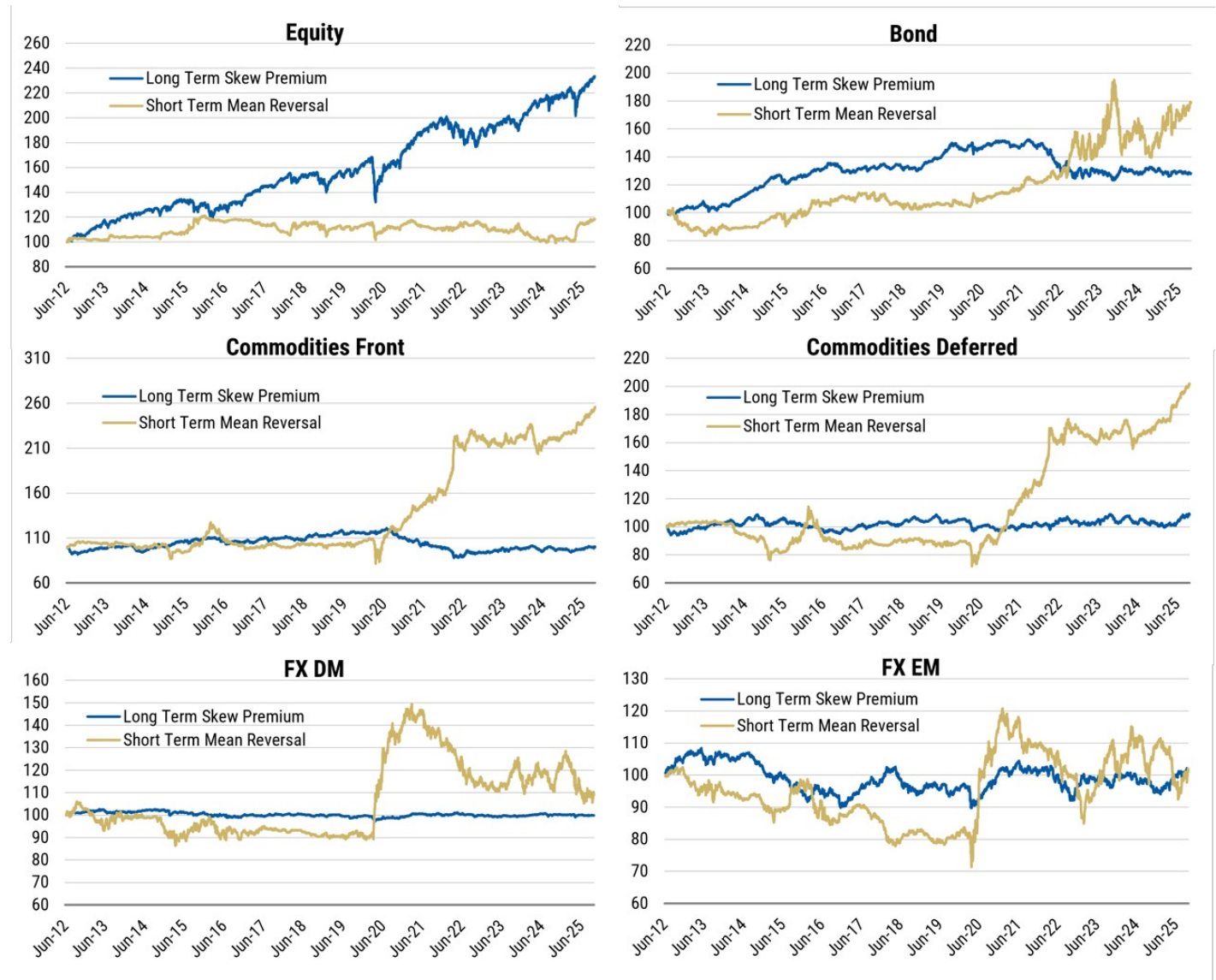
Next, we regress future returns on realized skewness and examine the t-statistics for statistical significance. The rolling 5-year t-statistics for the slope of this regression line are for bonds and equities around a median of -1.5. For commodities the median of the t-statistics for the link between skewness metric and subsequent performance is around -1.0. We show in [Exhibit 8](#) the evolution of the rolling t-stat over time, finding it to range from values around +/-0 to significantly negative (below -2.0) for all asset classes. Thus, there is a time-varying element to the link between measured skewness and future performance, where skewness matters in some periods and not in others. These results suggest that realized skewness generates a time-varying premium.

Exhibit 8: T-statistics of regressing forward returns on skewness

Source: Morgan Stanley Research; Note: The red line indicates the critical t-value of -1.282 for a one-tailed t-test at the 10% significance level ($\alpha = 0.10$).

We further explore this topic by decomposing return expectations associated with skewness into two components – a persistent long-term effect and a short-term tactical effect. The reason for this decomposition is that it allows us to distinguish between the effects of a longer-term risk premium linked to skewness and shorter-term tactical effects related to mean reversals. [Exhibit 9](#) illustrates the results of this analysis. To estimate the long-term skew premium, we fix portfolio weights using historical averages for each asset. For the short-term mean-reversion effect, we subtract the average weight from time-varying weights to isolate directional shifts.

Across asset classes, we observe an overall positive impact from both the long-term premium and the short-term reversal effect. The only exceptions are front-month commodity futures and developed market FX, where the long-term skewness premium appears largely neutral. Overall, both return drivers prove effective across asset classes. However, our findings reveal distinct patterns across asset classes. For equities, most of the outperformance is driven by the long-term skew premium, reflecting the predominantly negative skewness of equity returns. Bonds show contributions from both the long-term skew premium and short-term mean-reversion effects, although the long-term component has become more muted in recent periods. Commodities, on the other hand, derive part of returns from long-term skewness but most from mean-reversion dynamics, which have been particularly strong following the Covid period.

Exhibit 9: Long-term versus short-term skewness return decomposition

Source: Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

This section illustrated that realized skewness is reasonably consistent across return periods and estimation windows. We also show that average skewness across assets within an asset class generally revolves around zero within a rather narrow band. The number also does not exceed ± 1 frequently. Overall, we find that there is an economically relevant link between realized skewness and future asset performance. We use these insights to parametrize the algorithm behind our cross-asset Skewness strategy in the following section.

Portfolio Construction and Strategy Performance

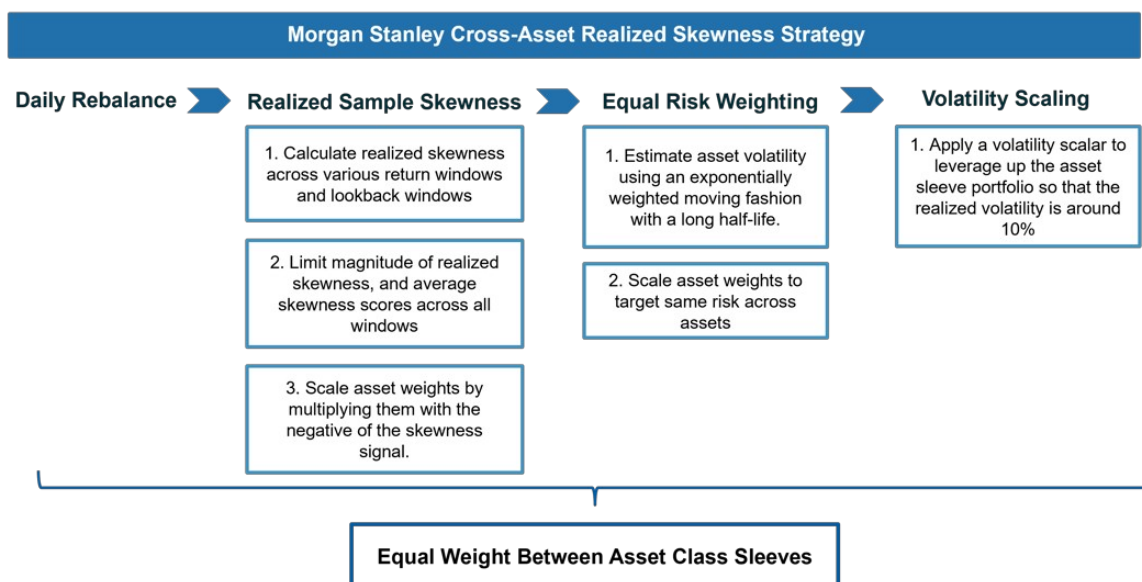
In this section, we give an overview of the construction of our strategy and discuss the key performance characteristics.

Our investment universe covers 109 contracts across equities, commodities, rates, and FX. A full list of assets is in [Overview of investment universe](#). Based on the analyses so far, we calculate skewness over (1) different estimation windows and (2) different return periods. This leads to multiple skewness metrics being calculated – 12 in total – which we average and cap/floor (at 1/-1, respectively). The position we then take in each asset is proportional to the negative of the average skewness. We scale each asset to a target volatility to make the exposures comparable within and across asset classes from a risk perspective.

This approach means that we follow a directional long/short portfolio construction approach. Given the results from the previous section, illustrating that skewness roughly revolves around zero, the strategy has a limited market exposure overall in the long run. However, it does exhibit time-varying leverage as well as time-varying market beta. This feature makes the strategy similar in style to Trend Following. As a matter of fact, we will see later that Skewness and Trend Following strategies have low correlations and are fundamentally diversifying to each other.

To limit the directionality in the Skewness strategy, we add a cap and floor on market beta to the strategy to ensure that it does not load up excessively on market risk. We show in [Robustness Tests](#) that the beta cap has only a limited impact on overall performance. See [Exhibit 10](#) for more details on the strategy.

Exhibit 10: From signal computation to the final weights



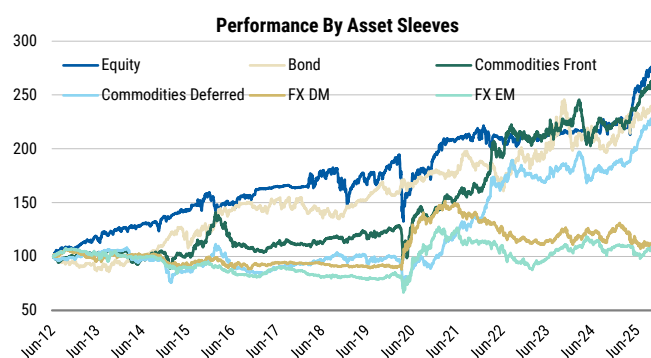
Source: Morgan Stanley Research

We show the cumulative performance of the asset class sleeves in [Exhibit 11](#), and performance of the overall strategy in [Exhibit 12](#). The strategy shows a persistent performance over our core backtesting period from 2012. In particular since the Covid sell-

off the strategy shows strong returns up until the more recent past. A drawdown in 2020 is evident, which is essentially a result of the skewness turning negative for many assets during the Covid sell-off and the strategy starting to go long, speculating on a market rebound, before such a rebound actually happened. While this positioning eventually paid off with very strong performance in late 2020 and early 2021, there is a small dip before the positioning rebounds.

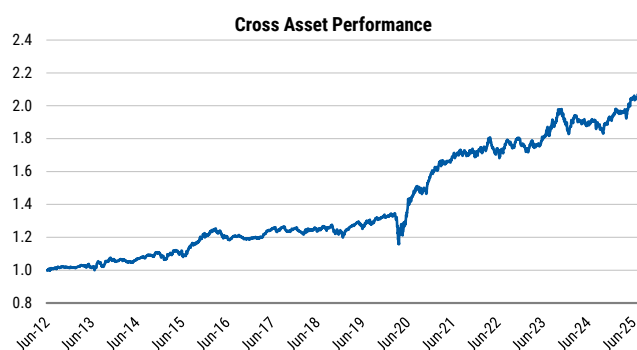
For the individual asset class sleeves, performance is strong in equities, bonds, and commodities. While we find also positive returns in the FX space, they are rather low. We attribute the comparatively low performance of the FX sleeves to the fact that FX exposures are always long one currency versus another, making it harder to determine whether market participants are more concerned about upside or downside risks and making the case of a risk premium less clear.

Exhibit 11: Strategy performance by asset



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 12: Top-level performance



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

In [Exhibit 13](#) we summarize the performance statistics associated with the cumulative return charts. Sharpe ratios in equities, bonds, and commodities are 0.6 or higher. In FX we are positive but close to zero in terms of Sharpe ratio. Not surprisingly, the skewness of the strategy is negative, which is also illustrated by ratios of maximum drawdown to volatility of -1.8 (for bonds) to -3.7 (for EM FX). When aggregating this information at the asset class level, we arrive at a rather attractive Sharpe ratio of 1.01 post-cost. The maximum drawdown-to-volatility ratio is -2.4, which is broadly consistent with other long/short strategies. For comparison, the Morgan Stanley Enhanced Trend Following strategy shows a ratio of -1.7, reflecting its more defensive profile.

Exhibit 13: Strategy performance statistics

	Equity	Bond	Commodities Front	Commodities Deferred	FX DM	FX EM	Cross Asset
CAGR	7.81%	6.68%	7.42%	6.27%	0.79%	0.59%	5.91%
Annualized Vol	10.39%	10.98%	11.28%	10.81%	9.65%	10.26%	5.88%
CAGR / Vol	0.75	0.61	0.66	0.58	0.08	0.06	1.01
Max DD	-31.80%	-20.10%	-31.65%	-35.33%	-29.23%	-38.04%	-13.95%
Max DD / Vol	-3.06	-1.83	-2.81	-3.27	-3.03	-3.71	-2.37
Calmar Ratio	0.25	0.33	0.23	0.18	0.03	0.02	0.42
Sortino Ratio	1.04	0.86	0.95	0.80	0.12	0.08	1.45
VaR (1%, Monthly)	-8.89%	-7.28%	-9.07%	-10.72%	-5.92%	-7.42%	-3.83%
CVaR (1%, Monthly)	-16.69%	-8.71%	-12.76%	-14.87%	-6.80%	-10.28%	-6.58%
Hit Rate (Monthly)	64.81%	59.83%	58.54%	59.11%	48.65%	48.27%	62.45%
Skewness	-1.01	-0.25	-0.43	-0.91	0.07	0.91	-0.41
Excess Kurtosis	29.79	3.03	20.08	15.24	6.27	23.26	15.89

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

This substantial improvement in Sharpe ratio at a multi-asset level is driven by the low correlations between our asset class sleeves. [Exhibit 14](#) shows that correlations are below 0.3 with two exceptions – the correlations among the two commodity and the two FX sub-sleeves, which are of course fundamentally linked. We provide a table with tickers summarizing the performance of the asset class sleeves and the overall cross-asset strategy performance in the Appendix ([Exhibit 64](#)).

Overall, the results illustrated a strong overall performance of our Skewness strategy. It works well in all major asset classes, with only FX showing weaker results. Combining the strategy across asset classes increases risk-adjusted performance through diversification benefits. The performance is persistent through time and shows only limited correlations to traditional markets.

Exhibit 14: Correlation between asset sleeves

Based on Daily Returns	Equity	Bond	Commodity Front	Commodity Deferred	FX DM	FX EM
Equity	1.00	-0.05	0.19	0.28	0.19	0.28
Bond	-0.05	1.00	0.06	0.03	0.08	0.05
Commodity Front	0.19	0.06	1.00	0.78	0.13	0.13
Commodity Deferred	0.28	0.03	0.78	1.00	0.17	0.18
FX DM	0.19	0.08	0.13	0.17	1.00	0.51
FX EM	0.28	0.05	0.13	0.18	0.51	1.00

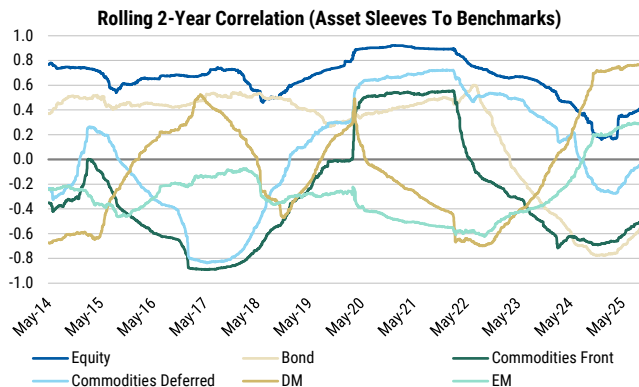
Source: Bloomberg, Morgan Stanley Research

The strategy permits directional exposures across individual asset classes while capping overall market beta at 30% prior to targeting volatility. Below, we outline the rationale behind this approach.

As illustrated in [Exhibit 58](#) in the Appendix, within equity and bond markets, assets frequently exhibit high correlation, which can cause skewness signals to move in tandem. By enabling directional positioning at the asset class level, the strategy inherently incorporates an element of market timing. However, by constraining the aggregate beta, we aim to isolate the skewness premium from broader market leverage effects – ensuring that performance attribution remains focused on the intended signal rather than unintended market exposure.

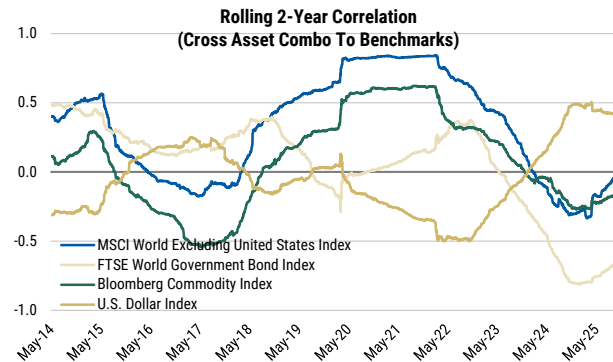
Exhibit 15 presents the rolling correlation of each asset sleeve relative to its respective market benchmark, while Exhibit 16 illustrates the correlation between the cross-asset portfolio and selected market benchmarks. The data reveal a wide range of correlation levels across markets, with values fluctuating between positive and negative territory.

Exhibit 15: Asset sleeves correlation to respective markets



Source: Bloomberg, Morgan Stanley Research

Exhibit 16: Cross-asset combination correlation to markets



Source: Bloomberg, Morgan Stanley Research

This section examines the strategy's design and its performance characteristics over the backtest period. We demonstrate that the skewness premium delivered strong risk-adjusted returns across most asset classes, with the exception of FX, highlighting its robustness in multi-asset contexts. The cross-asset portfolio achieved a Sharpe ratio of 1.01, benefiting from diversification effects that helped to smooth return volatility. In the next section, we place the Realized Skewness strategy alongside the Trend Following strategy and discuss the advantages of combining the two, highlighting how their complementary characteristics can enhance portfolio efficiency.

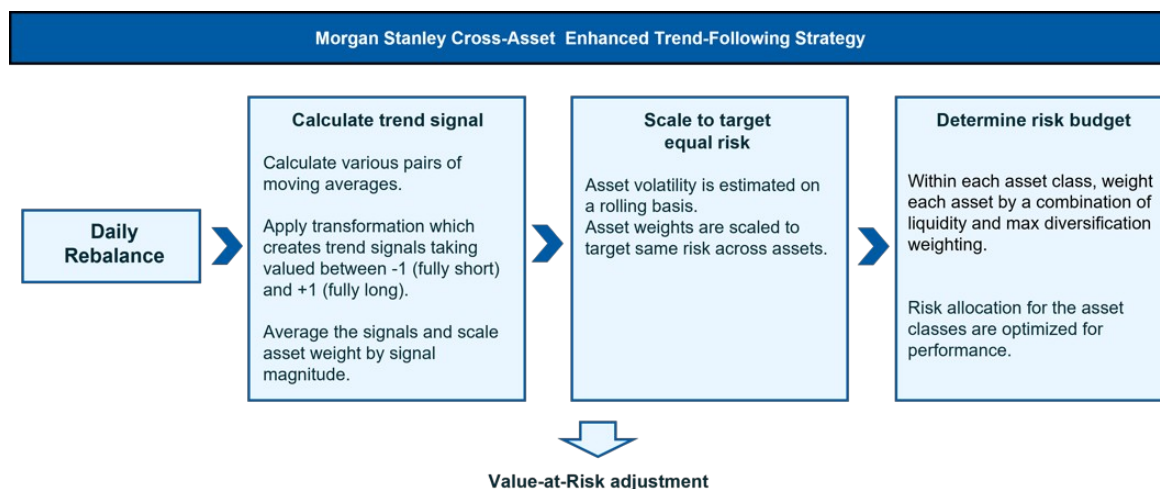
Interaction Between Skewness and Trend Following

In the preceding section, we examined the design and empirical performance of the Skewness strategy. Owing to its mean-reversion characteristics and directional construction, the strategy exhibits behavior that contrasts with the Trend Following approaches. This distinction makes it conceptually comparable to Trend Following strategies while offering complementary risk exposures. As a result, the Skewness strategy can serve as an effective diversifier when paired with Trend Following, enhancing portfolio robustness by mitigating concentration in directional bets and improving overall risk-adjusted returns.

Enhanced Trend Following

We introduced the Trend Following strategy in [Crisis Alpha Through Defensive Trend Following](#), April 28, 2020, which detailed the design of an algorithm modulated for enhanced crisis alpha characteristics. The Enhanced Trend Following strategy, as the name suggests, builds on the original design by incorporating asset-class specific adjustments and optimizing for maximum diversification and Sharpe ratio.

Exhibit 17: Enhanced Trend Following methodology



Source: Morgan Stanley Research

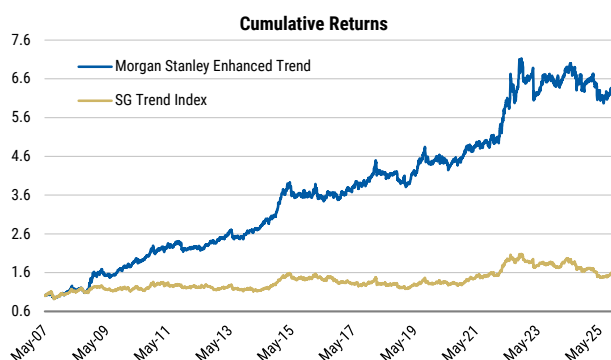
The Enhanced Trend Following strategy, which we compare to our Skewness strategy, shares several design characteristics. At a high level, both strategies are directional, resulting in time-varying market betas, and they take long or short positions based on signals derived from historical return dynamics across multiple horizons. These signals are capped and floored at ± 1 to mitigate outliers and are adjusted for asset volatility to maintain equal risk within asset classes. Both target a specific volatility budget at an asset level.

However, the signals themselves differ and often diverge. Trend signals compare fast-moving return averages with slower ones, going long when short-term returns exceed long-term returns and vice versa. This approach favors assets with recent positive performance. In contrast, the Skewness strategy overweights assets that have recently underperformed. By design, the two strategies behave differently and tend to

complement each other, particularly during periods when signals are stretched.

In [Exhibit 18](#), we show the historical performance of the Enhanced Trend strategy, with its performance statistics summarized in [Exhibit 19](#).¹ The strategy has delivered consistent positive returns, achieving a Sharpe ratio of 0.94. It is notably defensive during crisis periods, with a maximum drawdown to volatility ratio of only -1.46. Its performance also significantly exceeds that of the SG Trend Index benchmark, which recorded a Sharpe ratio of just 0.23 over the same backtest period. We believe that Enhanced Trend has strong potential to offset losses from the Skewness strategy when it pivots to mean reversion during market drawdowns. Conversely, Skewness can help to capture market rebounds when Enhanced Trend adjusts its positions.

Exhibit 18: Enhanced Trend performance



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2007/05/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 19: Performance statistics of Enhanced Trend

	Morgan Stanley Enhanced Trend	SG Trend Index
CAGR	10.42%	2.58%
Annualized Vol	11.11%	11.12%
CAGR / Vol	0.94	0.23
Max DD	-16.22%	-29.62%
Max DD / Vol	-1.46	-2.66
Calmar Ratio	0.64	0.09
Sortino Ratio	1.34	0.31
VaR (1%, Monthly)	-7.76%	-8.44%
CVaR (1%, Monthly)	-9.61%	-10.64%
Hit Rate (Monthly)	62.88%	55.58%
Skewness	-0.20	-0.76
Excess Kurtosis	6.63	3.58

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2007/05/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

To enable a fair comparison, we extend the backtest period of the Skewness strategy so that it begins on the same start date as the Enhanced Trend Following strategy. For further details on this iteration, see [Extending backtest history](#) in [Robustness Tests](#). It is worthwhile to highlight that we need to be careful in interpreting the early years of this extended backtest as the investment universe starts to become rather concentrated due to fewer assets being available for investment at the time.

Correlation between Realized Skewness and Enhanced Trend Following

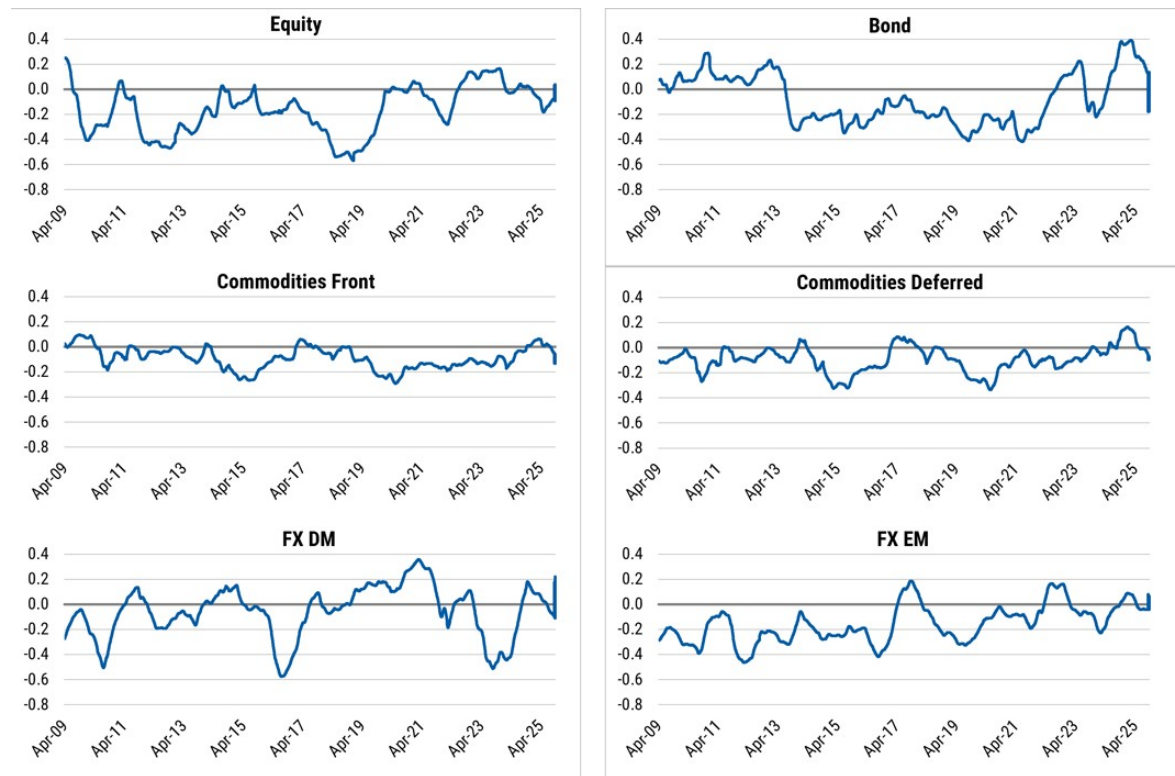
We begin by examining the correlation between the two strategies at the weights level. [Exhibit 20](#) reports the average rolling two-year correlation of target weights across asset classes. For each asset traded in both strategies, we calculate its rolling correlation and then average these values within each asset class. Historically, correlations have remained very low. The maximum observed correlation was 0.4, which was short-lived, while most of the time correlations have been neutral to negative, reaching lows of nearly -0.6. Correlation scores for commodities are particularly constrained around zero.

This low and often negative correlation suggests that the Skewness strategy and the Enhanced Trend Following strategy capture distinct sources of return. Their differing

1. Note that we show the performance of the SG Trend Index here in excess of the risk-free rate.

exposures imply strong diversification benefits when combined in a multi-strategy portfolio, as periods of underperformance in one approach are less likely to coincide with similar outcomes in the other.

Exhibit 20: Rolling two-year correlation of target weights

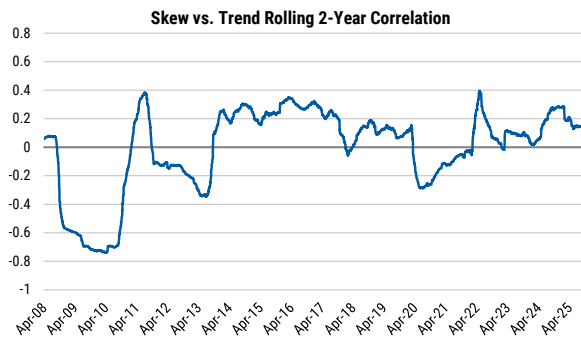


Source: Morgan Stanley Research

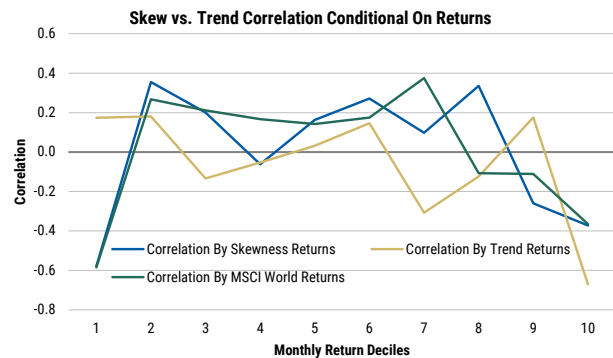
[Exhibit 21](#) illustrates the correlation at the return level. Once again, we observe limited correlation throughout the sample period. Notably, correlations turn negative during both the 2008 and 2020 crisis episodes, indicating the complementary nature of the two strategies.

In [Exhibit 22](#), we begin by sorting historical monthly returns into ten decile buckets based on MSCI World returns and then compute the correlation between Skewness and Trend returns within each bucket. We repeat this analysis for Trend Following and Skewness as the conditioning factor in the deciling.

When grouping by returns of Skewness, Trend, or the MSCI World Index, we observe that conditional correlations are particularly negative in periods when equities or Skewness underperform (i.e., decile 1), indicating a strong benefit from diversification in portfolios that contain both strategies. When concentrating on periods where a strategy or the market outperform (i.e., decile 10), correlations between Skewness and Trend are also negative indicating a diversification effect. Thus, in time periods where the two strategies or the market show an extreme performance there is a tendency to neutralize exposures, while in other periods we find muted correlations around 0. This effect has a tendency to remove tail events in the combined portfolio and lead to higher risk-adjusted returns in the belly of the distribution.

Exhibit 21: Rolling 2-year correlation of returns

Source: Morgan Stanley Research

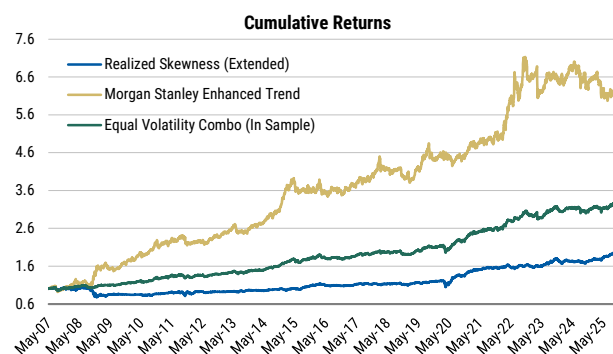
Exhibit 22: Correlation conditional on returns

Source: Morgan Stanley Research

Combining Realized Skewness and Enhanced Trend Following

Next, we combine the two strategies and evaluate the resulting performance and potential improvements in risk-adjusted returns. Each strategy is allocated an equal risk budget, and inverse-volatility weights are computed using in-sample volatility estimates. [Exhibit 23](#) compares the performance of the combined portfolio against the individual strategies, while [Exhibit 24](#) summarizes key performance statistics.

As expected, the combined approach delivers superior results. The Sharpe ratio rises to 1.15, outperforming both stand-alone strategies. Maximum drawdown is significantly reduced and, after volatility adjustment, it stands at -1.7. Additionally, the hit rate – defined as the percentage of months with positive returns – improves to 66.52%, indicating a more consistent return profile. These findings highlight the benefits of diversification across complementary signals. By blending Skewness and Trend Following exposures, the portfolio achieves higher efficiency, mitigates tail risk, and enhances stability across market regimes.

Exhibit 23: Skewness and Trend performance

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2007/05/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 24: Performance statistics

	Realized Skewness (Extended)	Enhanced Trend	Equal Volatility Combo (In Sample)
CAGR	3.72%	10.42%	6.63%
Annualized Vol	7.38%	11.11%	5.75%
CAGR / Vol	0.50	0.94	1.15
Max DD	-25.22%	-16.22%	-9.77%
Max DD / Vol	-3.42	-1.46	-1.70
Calmar Ratio	0.15	0.64	0.68
Sortino Ratio	0.70	1.34	1.60
VaR (1%, Monthly)	-6.70%	-7.76%	-4.59%
CVaR (1%, Monthly)	-9.83%	-9.61%	-6.11%
Hit Rate (Monthly)	59.24%	62.88%	66.52%
Skewness	-0.43	-0.20	-0.73
Excess Kurtosis	18.47	6.63	4.48

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2007/05/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

[Exhibit 25](#) illustrates how combining Skewness and Trend Following reshapes the distribution of monthly returns, as shown through histograms.² Visually, the combination

2. Strategy returns are scaled to a 10% volatility level for comparability.

reduces kurtosis, indicating a less extreme distribution compared to the individual strategies. Overall, the median return of the combined approach exceeds that of Skewness and Trend Following on their own, suggesting an improved central tendency. Importantly, the left tail risk inherent in Skewness – representing large negative shocks – is significantly truncated when Trend Following is added. At the same time, the right tail of outsized positive shocks is moderated, trading some upside potential for a more stable and consistent return profile.

Exhibit 25: Distribution of strategy returns



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2007/05/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Performance during market distress

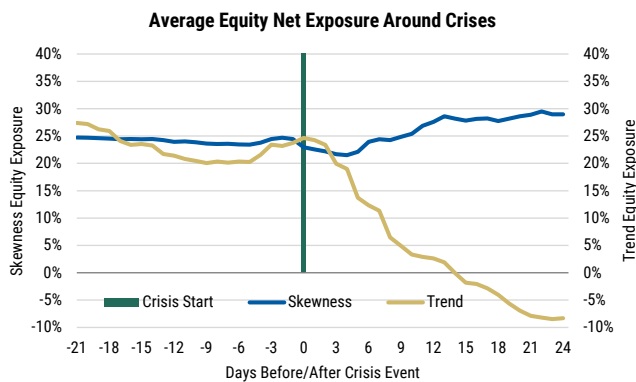
We focus on specific historical periods of major market dislocations to better understand how the two strategies respond under stress. [Exhibit 26](#) illustrates the average aggregate net equity exposure before and after equity drawdowns, while [Exhibit 27](#) shows exposures around the subsequent rebounds.³ The x-axis represents days relative to the start of the drawdown or rebound, with day 0 marking the onset of the crisis or recovery (highlighted by the green bar).

During crises, Trend reacts swiftly by reducing equity exposure and eventually moving

3. The crisis periods considered include the global financial crisis in 2008 (from the Lehman collapse to the Fed funds trough), 4Q18, the Covid shock, and January-June 2022.

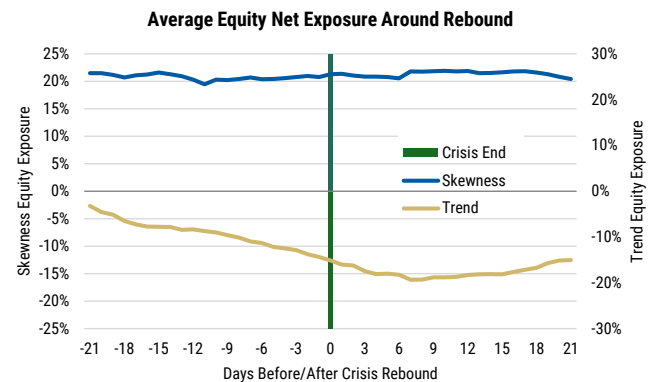
short as downward momentum strengthens. In contrast, Skewness tends to increase leverage gradually, reflecting its sensitivity to tail-risk shocks. When rebounds occur, Skewness has typically already added positions to capture the reversal, with exposures plateauing and then declining slowly as negative skewness normalizes. Meanwhile, Trend remains defensive at the rebound point, with positions below zero, and requires time to rebuild exposure as upward momentum re-emerges.

Exhibit 26: Behavior around crises



Source: Bloomberg, Morgan Stanley Research

Exhibit 27: Behavior around rebound



Source: Bloomberg, Morgan Stanley Research

This analysis highlights the structural differences in behavior: Trend prioritizes momentum and risk reduction during stress, whereas Skewness seeks to exploit dislocations and reversal opportunities. Understanding these dynamics is critical for portfolio construction, as combining the two can provide complementary responses across market regimes. We also include a more detailed analysis of asset-level behavior during a major crisis in [Skewness versus trend weights revolution during specific events](#) in the Appendix.

After examining the evolution of portfolio weights during crisis periods, we turn to the analysis of return outcomes. One way to identify distressed periods is by examining the worst-performing months of the MSCI World Index. [Exhibit 28](#) highlights the 10 most severe monthly drawdowns and compares the returns of the individual strategies and their combination. To assess resilience, we also analyze the first positive month following each drawdown to understand how quickly the strategies recover.

During these stress periods, the Realized Skewness strategy tends to be more exposed to risk events, often turning negative when markets decline. In contrast, the Enhanced Trend strategy demonstrates defensive characteristics, generally delivering positive returns during these months. When combined, the portfolio exhibits a more balanced profile – on average, flat returns during the worst months, mitigating downside risk.

Recovery dynamics are particularly noteworthy. Realized Skewness rebounds strongly, with all 10 months positive during the post-drawdown phase. Enhanced Trend, however, tends to give back some gains during market recoveries. The combined portfolio benefits from this complementary behavior, achieving more consistent positive returns during rebound periods. When combining the worst months and the recovery months together, the combined portfolio now achieves more balanced returns at +0.91% (i.e., -0.58% + 1.49%).

Exhibit 28: Performance during and after MSCI World worst months

MSCI World Worst Months					MSCI World First Positive Months After Worst Months				
Worst Month	MSCI World Return	Skewness Return	Trend Return	Skew + Trend Return	First Positive Month After	MSCI World Return	Skewness Return	Trend Return	Skew + Trend Return
2008-10-31	-18.96%	-10.33%	20.90%	2.12%	2008-12-31	3.21%	2.14%	1.97%	2.39%
2020-03-31	-13.23%	-5.00%	1.67%	-2.34%	2020-04-30	10.92%	3.97%	-1.14%	2.07%
2008-09-30	-11.89%	-4.89%	5.47%	-0.76%	2008-12-31	3.21%	2.14%	1.97%	2.39%
2009-02-28	-10.24%	-0.88%	3.07%	0.70%	2009-03-31	7.54%	2.12%	-1.52%	0.77%
2010-05-31	-9.58%	-1.18%	-3.43%	-2.08%	2010-07-31	8.11%	1.93%	2.20%	2.04%
2022-09-30	-9.30%	-2.04%	10.20%	2.82%	2022-10-31	7.18%	2.97%	-0.83%	1.39%
2009-01-31	-8.76%	-2.18%	2.17%	-0.45%	2009-03-31	7.54%	2.12%	-1.52%	0.77%
2022-06-30	-8.66%	-0.48%	8.16%	2.93%	2022-07-31	7.94%	2.09%	-5.82%	-1.12%
2011-09-30	-8.64%	-9.82%	-0.46%	-6.10%	2011-10-31	10.34%	10.46%	-6.34%	3.95%
2012-05-31	-8.63%	-3.13%	-1.78%	-2.59%	2012-06-30	5.10%	1.07%	-1.10%	0.21%
Average	-10.79%	-3.99%	4.60%	-0.58%	Average	7.11%	3.10%	-1.21%	1.49%

Source: Bloomberg, Morgan Stanley Research

Exhibit 29 highlights the 10 worst-performing months for the Realized Skewness and Enhanced Trend strategies. During these specific drawdown periods, we observe that the other strategy typically delivers positive returns on average, providing a natural offset. While the combined portfolio does not always achieve positive returns in these extreme months, the overall risk is reduced significantly compared to either strategy in isolation.

Exhibit 29: Performance during and after strategies' worst months

Realized Skewness 10 Worst Months					Realized Skewness First Positive Months After Worst Months				
Worst Month	Skewness Return	Trend Return	Skew + Trend Return	MSCI World Return	First Positive Month After	Skewness Return	Trend Return	Skew + Trend Return	MSCI World Return
2008-10-31	-10.33%	20.90%	2.12%	-18.96%	2008-12-31	2.14%	1.97%	2.39%	3.21%
2011-09-30	-9.82%	-0.46%	-6.10%	-8.64%	2011-10-31	10.46%	-6.34%	3.95%	10.34%
2020-03-31	-5.00%	1.67%	-2.34%	-13.23%	2020-04-30	3.97%	-1.14%	2.07%	10.92%
2008-09-30	-4.89%	5.47%	-0.76%	-11.89%	2008-12-31	2.14%	1.97%	2.39%	3.21%
2023-11-30	-4.23%	-2.99%	-3.73%	9.38%	2024-01-31	3.25%	-2.34%	1.01%	1.20%
2022-04-30	-4.18%	5.66%	-0.27%	-8.31%	2022-05-31	0.42%	-2.20%	-0.63%	0.08%
2019-05-31	-3.41%	-0.18%	-2.12%	-5.77%	2019-06-30	2.80%	4.83%	3.61%	6.59%
2016-04-30	-3.29%	-1.16%	-2.44%	1.58%	2016-05-31	1.09%	-1.78%	-0.05%	0.56%
2023-12-31	-3.24%	3.21%	-0.67%	4.91%	2024-01-31	3.25%	-2.34%	1.01%	1.20%
2012-05-31	-3.13%	-1.78%	-2.59%	-8.63%	2012-06-30	1.07%	-1.10%	0.21%	5.10%
Average	-5.15%	3.03%	-1.89%	-5.96%	Average	3.06%	-0.85%	1.60%	4.24%

Enhanced Trend 10 Worst Months					Enhanced Trend First Positive Months After Worst Months				
Worst Month	Skewness Return	Trend Return	Skew + Trend Return	MSCI World Return	First Positive Month After	Skewness Return	Trend Return	Skew + Trend Return	MSCI World Return
2023-03-31	-1.19%	-8.53%	-4.11%	3.09%	2023-04-30	0.15%	1.29%	0.60%	1.75%
2011-10-31	10.46%	-6.34%	3.95%	10.34%	2011-11-30	-2.82%	0.45%	-1.72%	-2.44%
2008-07-31	-0.06%	-5.97%	-2.41%	-2.44%	2008-09-30	-4.89%	5.47%	-0.76%	-11.89%
2022-07-31	2.09%	-5.82%	-1.12%	7.94%	2022-08-31	0.74%	5.65%	2.70%	-4.18%
2024-10-31	3.03%	-5.68%	-0.46%	-1.98%	2024-11-30	1.31%	3.62%	2.22%	4.59%
2010-11-30	-0.16%	-5.66%	-2.36%	-2.16%	2010-12-31	1.67%	6.38%	3.53%	7.35%
2018-02-28	1.04%	-5.37%	-1.52%	-4.14%	2018-04-30	0.04%	0.23%	0.11%	1.15%
2008-04-30	3.82%	-5.30%	0.13%	5.26%	2008-05-31	1.64%	3.29%	2.30%	1.52%
2019-09-30	2.12%	-5.17%	-0.80%	2.13%	2019-11-30	1.07%	0.40%	0.80%	2.79%
2022-11-30	-0.50%	-4.94%	-2.27%	6.95%	2022-12-31	-1.36%	1.48%	-0.23%	-4.25%
Average	2.07%	-5.88%	-1.10%	2.50%	Average	-0.24%	2.83%	0.96%	-0.36%

Source: Bloomberg, Morgan Stanley Research

In the subsequent recovery periods, the interaction between the two strategies becomes more nuanced. Due to their low to negative correlations, the non-drawdown strategy can occasionally post negative returns during rebounds. However, the combined portfolio still demonstrates strong recovery dynamics, benefiting from diversification and complementary performance drivers. These findings show that even when individual components experience stress, the combined approach mitigates tail risk and builds a resilient return profile over time.

We next focus on the performance of the stand-alone strategies and the combined

portfolio during historical periods of market distress. [Exhibit 30](#) summarizes returns across major crisis episodes since 2007. The defensiveness of the Enhanced Trend strategy notably strengthened outcomes during severe stress events – most prominently the 2008 global financial crisis and the first half of 2022, when markets experienced prolonged volatility. In these periods, the combined portfolio delivered significantly positive returns, underscoring the value of incorporating Trend Following signals for downside protection.

Conversely, there were episodes where Trend underperformed, such as the 2013 taper tantrum, the 2018 VIX crash, and the sharp drawdown during Liberation Day. In these cases, losses from the Realized Skewness strategy were relatively contained, helping to cushion overall portfolio drawdowns. This complementary behavior highlights the structural diversification benefits of combining strategies with distinct sensitivities to market regimes. In early recoveries, when Trend Following may still be short in equity markets after the drawdown, Skewness shows particularly strong performance, leading in the combined portfolio to a mean return of +1.47%. Thus, combining Trend Following with Skewness signals allows investors to benefit from market recovery more than Trend portfolios alone. Across the drawdown and the following six months of recovery, we achieve a return of +4.75% (i.e., -1.20% + 5.95%).

Exhibit 30: Performance during crisis periods

Events	Start	End	Realized Skewness (Extended)	Enhanced Trend	Equal Volatility Combo (In Sample)
GFC 2008 (DJIA Top - Lehman Collapse)	2007-07-19	2008-09-19	-4.68%	2.69%	-1.49%
GFC 2008 (Lehman Collapse - Fed Fund Trough)	2008-09-19	2008-12-16	-14.14%	40.93%	6.33%
GFC 2008 (Fed Fund Trough - Global Stimulus)	2008-12-16	2009-04-02	1.24%	0.04%	0.96%
GFC 2008 (Global Stimulus - Year End)	2009-04-02	2009-12-31	-0.13%	11.03%	4.34%
GFC 2008	2007-07-19	2009-04-02	-17.14%	44.78%	5.76%
Flash Crash 2010	2010-05-05	2010-05-25	-2.18%	-2.59%	-2.34%
Greek Debt Crisis 2011	2011-07-22	2011-08-09	-3.39%	-0.12%	-2.08%
Taper Tantrum 2013	2013-05-22	2013-06-24	-1.66%	-8.75%	-4.56%
Chinese Stock Market Turbulence 2015	2015-08-17	2015-08-25	0.51%	-2.20%	-0.57%
VIX Crash 2018	2018-01-26	2018-02-09	-0.78%	-8.78%	-4.00%
Q4 2018	2018-10-03	2018-12-24	-5.14%	-3.34%	-4.35%
COVID-19	2020-02-19	2020-03-24	-10.25%	-2.67%	-7.20%
January-June 2022	2022-01-01	2022-06-16	-2.55%	31.43%	10.21%
Trump Liberation Day	2025-04-02	2025-04-08	-1.24%	-5.37%	-2.89%
Crisis Returns Mean			-4.38%	4.24%	-1.20%
1M After Crisis Mean			2.12%	0.55%	1.47%
6M After Crisis Mean			7.19%	4.08%	5.95%

Source: Bloomberg, Morgan Stanley Research

The performance analysis during crisis periods reinforces the findings from [Exhibit 26](#) and [Exhibit 27](#). Realized Skewness typically underperforms during market distress as it positions for mean reversion, while Trend adopts a risk-off stance and provides protection. However, when markets rebound, Skewness outperforms Trend due to its positive exposures. Combining the two strategies helps to mitigate losses during crises while preserving responsiveness during recoveries.

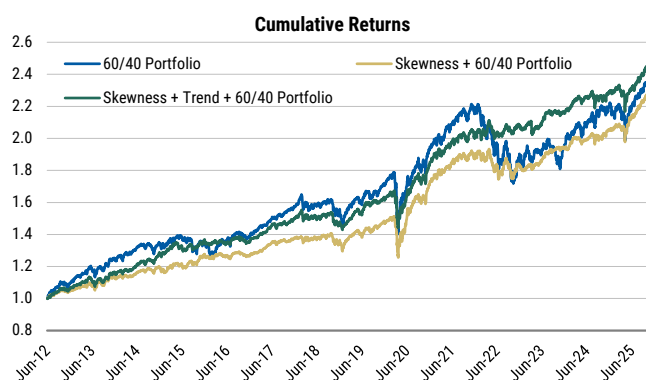
Adding to a traditional 60/40 portfolio

We further examine the impact of incorporating Skewness and Trend Following strategies into a traditional 60/40 portfolio. The baseline 60/40 allocation consists of the MSCI

World Index for equities and the Bloomberg US Aggregate Bond Index for fixed income. [Exhibit 31](#) and [Exhibit 32](#) present the cumulative return profiles and key performance statistics.

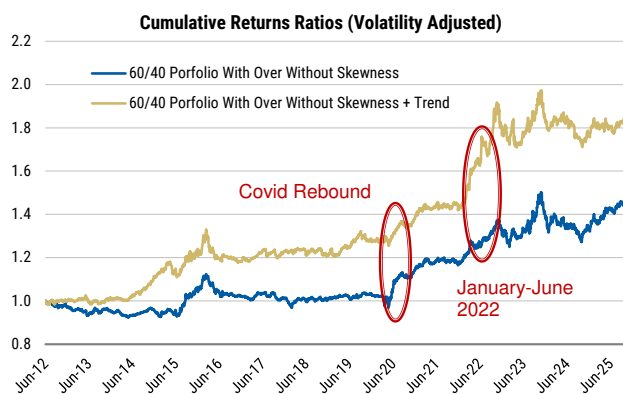
On its own, the 60/40 portfolio delivered a Sharpe ratio of 0.76 and a maximum drawdown-to-volatility ratio of -2.6, reflecting moderate risk-adjusted returns and vulnerability during stress periods. When we introduce the Skewness strategy with an equal risk weighting, the Sharpe Ratio improves from 0.76 to 1.07, indicating diversification benefits. More notable enhancement occurs when Trend Following is added alongside Skewness. Under this combined allocation, the Sharpe ratio rises substantially to 1.25.

Exhibit 31: Adding Skewness/Trend strategy to 60/40 portfolio performance



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 33: Cumulative performance ratio of 60/40 portfolio with versus without Skewness and Trend



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 32: Performance statistics when adding Skewness/Trend strategy to 60/40 portfolio

	60/40 Portfolio	Skewness + 60/40 Portfolio	Skewness + Trend + 60/40 Portfolio
CAGR	6.52%	6.26%	6.85%
Annualized Vol	8.58%	5.88%	5.49%
CAGR / Vol	0.76	1.07	1.25
Max DD	-22.33%	-17.03%	-13.84%
Max DD / Vol	-2.60	-2.90	-2.52
Calmar Ratio	0.29	0.37	0.50
Sortino Ratio	1.06	1.50	1.73
VaR (1%, Monthly)	-6.93%	-4.54%	-3.73%
CVaR (1%, Monthly)	-11.22%	-8.20%	-6.94%
Hit Rate (Monthly)	65.99%	66.56%	68.66%
Skewness	-0.73	-0.83	-1.03
Excess Kurtosis	13.92	22.37	9.84

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

In [Exhibit 33](#), we present the cumulative return ratios obtained by dividing the cumulative returns of a 60/40 portfolio enhanced with Trend and Skewness strategies by those of a traditional 60/40 portfolio, after adjusting both to the same volatility level. This normalization ensures a fair comparison of performance across strategies. The results indicate that the augmented portfolio consistently outperforms the traditional allocation throughout the backtest period, highlighting the persistent benefit of incorporating Trend and Skewness into a diversified portfolio.

Adding Skewness enhances the ability to capture mean reversion during market rebounds. For example, the combined portfolio outperformed the traditional 60/40 allocation during the recovery following the Covid drawdown. We also show that during prolonged periods of distress, such as January to June 2022, incorporating both Skewness and Trend improved returns significantly. For a detailed analysis of performance during crisis periods, see [Add Skewness and Trend to a 60/40 portfolio – crises periods analysis](#) in the Appendix.

Taken together, these results demonstrate that a multi-strategy approach not only enhances long-term risk-adjusted returns but also improves resilience during both systemic crises and idiosyncratic shocks. For portfolio construction, this suggests that allocating across strategies with differentiated drivers can reduce tail risk materially and deliver more stable performance through varying macro environments.

Robustness Tests

After discussing the performance of the Realized Skewness strategy both as a stand-alone approach and in combination with Trend Following, we now turn to a comprehensive suite of stability tests designed to validate the robustness of the strategy. In this section, we focus on examining key parameterization choices and evaluating the sensitivity of performance outcomes to variations in these inputs. This analysis ensures that the strategy's efficacy is not overly dependent on specific assumptions and remains resilient across different configurations.

Various skewness measures

There are several ways to measure skewness, each capturing different aspects of a distribution's asymmetry. In this section, we begin by exploring alternative skewness measures and examining how strategy performance varies depending on the chosen definition of sample skewness. In addition to the standard skewness formula we adopt, we consider two alternative definitions: quartile skewness and median skewness. The formulas for all three measures are presented below:

$$1. \text{ Quartile Skewness} = \frac{(Q3 - Q2) - (Q2 - Q1)}{Q3 - Q1},$$

where $Q1$ is the first Quartile, $Q2$ is the median,
and $Q3$ is the third Quartile,

$$2. \text{ Median Skewness} = \frac{3 \times (\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

$$3. \text{ Standard Skewness} = \frac{\sqrt{n(n-1)}}{n-2} \times \frac{\frac{1}{n} \times \sum_{i=0}^{n-1} (r_i - \bar{r})^3}{\left[\frac{1}{n} \times \sum_{i=0}^{n-1} (r_i - \bar{r})^2 \right]^{3/2}}$$

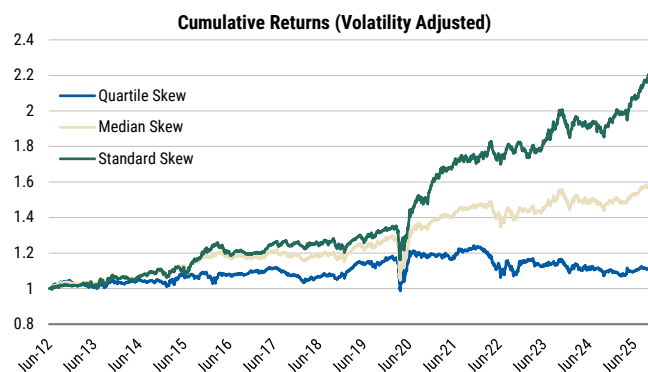
The key differences among these definitions lie in how they capture the behavior of distribution tails. Quartile skewness focuses on the relative distance between the median and the upper and lower quartiles, while median skewness compares the mean and the median. Both use rank-based or percentile-based statistics, making them robust to outliers and less sensitive to extreme values.

However, in real-world financial markets, return distributions often exhibit outliers – extreme events that investors care deeply about. These tail events can have a disproportionate impact on portfolio performance and risk perception. Therefore, while robust measures may offer stability, they may also under-represent the influence of rare but significant market moves. Capturing these extremes is crucial for strategies that aim to respond to tail risk.

In [Exhibit 34](#) and [Exhibit 35](#), we present the cumulative returns of strategies constructed using each of the three skewness definitions, along with their respective performance statistics. The strategy based on quartile skewness performs the worst, with returns remaining nearly flat over the historical period. The median skewness approach also underperforms relative to the standard skewness measure, delivering consistently lower

returns across all asset sleeves. These results highlight the importance of selecting a skewness definition that aligns with the objective of capturing meaningful asymmetries in return distributions – particularly those driven by extreme market events.

Exhibit 34: Performance by skewness measure



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 35: Performance statistics by skewness measure

	Quartile Skew	Median Skew	Standard Skew
CAGR	0.15%	1.95%	5.91%
Annualized Vol	0.94%	3.31%	5.88%
CAGR / Vol	0.16	0.59	1.01
Max DD	-2.77%	-11.16%	-13.95%
Max DD / Vol	-2.94	-3.38	-2.37
Calmar Ratio	0.05	0.17	0.42
Sortino Ratio	0.22	0.81	1.45
VaR (1%, Monthly)	-0.68%	-2.16%	-3.83%
CVaR (1%, Monthly)	-1.23%	-5.04%	-6.58%
Hit Rate (Monthly)	51.44%	60.70%	62.45%
Skewness	-0.68	-1.22	-0.41
Excess Kurtosis	23.46	37.83	15.89

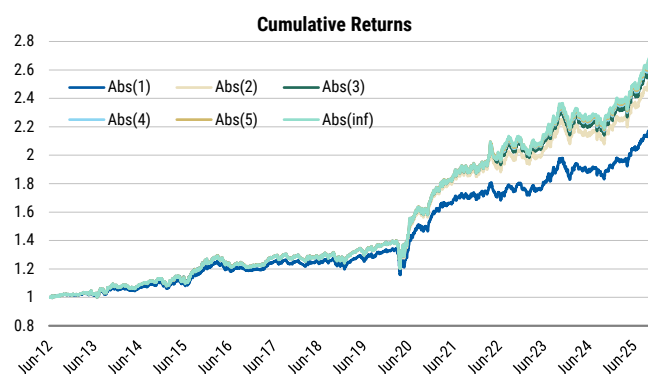
Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Signal capping and flooring

We explore the impact of capping the Realized Skewness signal at varying magnitudes. In our final design, we cap skewness at ± 1 for two key reasons. First, it provides a clear upper bound on asset-level exposure – ensuring that the maximum allocation to any single asset in the target portfolio does not exceed 100% prior to applying inverse-volatility weighting and volatility scaling. Second, it helps to filter out extreme signal values that often arise during market dislocations, where signal efficacy tends to deteriorate.

Exhibit 36 and Exhibit 37 present the strategy's performance under different skewness cap settings. While allowing higher or lower skewness values does lead to an increase in risk-adjusted returns, the improvement is largely concentrated in specific periods – most notably during the Covid shock and the 2022 market dislocation. These results suggest that while loosening the cap may enhance returns in isolated episodes, it also introduces greater sensitivity to tail events.

Exhibit 36: Performance by skewness cap and floor



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

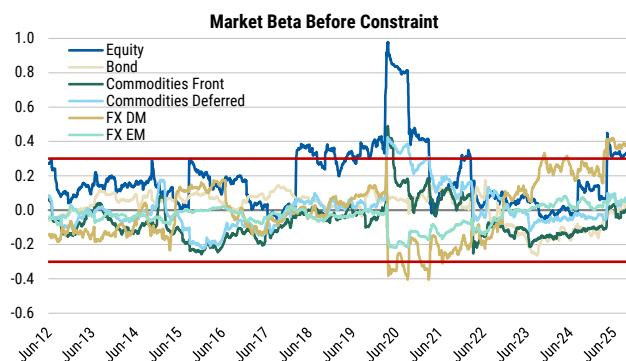
Exhibit 37: Performance statistics by skewness cap and floor

	Abs(1)	Abs(2)	Abs(3)	Abs(4)	Abs(5)	Abs(inf)
CAGR	5.91%	7.08%	7.36%	7.49%	7.53%	7.57%
Annualized Vol	5.88%	6.47%	6.60%	6.65%	6.68%	6.74%
CAGR / Vol	1.01	1.09	1.12	1.13	1.13	1.12
Max DD	-13.95%	-13.98%	-13.82%	-13.80%	-13.80%	-13.80%
Max DD / Vol	-2.37	-2.16	-2.09	-2.08	-2.07	-2.05
Calmar Ratio	0.42	0.51	0.53	0.54	0.55	0.55
Sortino Ratio	1.45	1.59	1.63	1.64	1.65	1.63
VaR (1%, Monthly)	-3.83%	-3.86%	-3.81%	-3.68%	-3.64%	-3.75%
CVaR (1%, Monthly)	-6.58%	-6.44%	-6.26%	-6.23%	-6.23%	-6.25%
Hit Rate (Monthly)	62.45%	63.28%	63.51%	63.40%	63.37%	63.31%
Skewness	-0.41	-0.24	-0.21	-0.20	-0.21	-0.24
Excess Kurtosis	15.89	16.10	16.53	16.40	16.20	15.87

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Market beta exposures

Exhibit 38: Beta before constraint



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30.

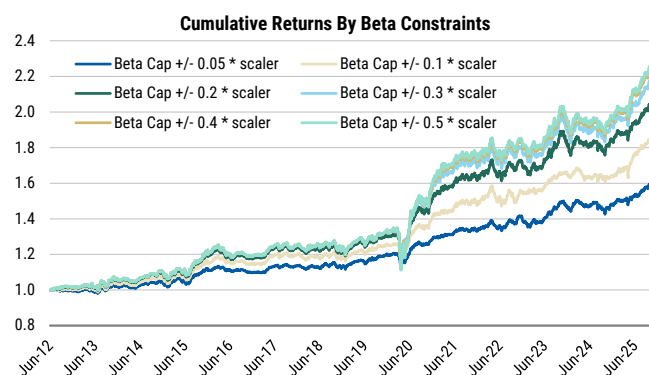
The Realized Skewness strategy allows each asset sleeve to take on market exposure up to $\pm 30\%$ prior to the application of volatility scaling. We examine the impact of adjusting this beta cap/floor at the cross-asset portfolio level. In [Exhibit 38](#), we show the market beta loadings of each asset class portfolio before applying any constraint. Most of time the beta exposures are within $\pm 30\%$ except during the Covid period when Equity beta spiked. Further restricting the beta constraint will require the portfolio reduce its reactivity to the changes in the skewness signal. Thus, constraining market beta at $\pm 30\%$ allows for the imposition of a cap on directional market risk without impacting the positioning of the strategy significantly.

In [Exhibit 39](#) and [Exhibit 40](#), our analysis shows that the Sharpe ratio peaks when the beta cap is set at 10%, and then declines gradually as the cap is either increased or decreased. While

cumulative returns tend to fall as market exposure is further restricted, the effect on risk-adjusted performance remains marginal. For our baseline strategy we pick the beta of $\pm 30\%$ as it strikes a good balance between performance, tail risks and minimal impact on weights through the beta constraint ([Exhibit 38](#)).

This suggests that moderate beta constraints strike a balance between capturing skewness-driven alpha and limiting unintended market leverage, without materially compromising the strategy's efficiency. For investors seeking a more market-neutral implementation to isolate the skewness premium, a beta cap of 5% emerges as a viable solution, delivering comparable risk-adjusted returns with minimal sacrifice in overall performance.

Exhibit 39: Performance by beta constraints



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 40: Performance statistics by beta constraints

	0.05 *	0.1 *	0.2 *	0.3 *	0.4 *	0.5 *
	scaler	scaler	scaler	scaler	scaler	scaler
CAGR	3.51%	4.63%	5.44%	5.91%	6.15%	6.21%
Annualized Vol	3.66%	4.44%	5.31%	5.88%	6.31%	6.55%
CAGR / Vol	0.96	1.04	1.02	1.01	0.97	0.95
Max DD	-7.47%	-8.56%	-11.23%	-13.95%	-16.12%	-17.62%
Max DD / Vol	-2.04	-1.93	-2.12	-2.37	-2.56	-2.69
Calmar Ratio	0.47	0.54	0.48	0.42	0.38	0.35
Sortino Ratio	1.40	1.53	1.49	1.45	1.39	1.35
VaR (1%, Monthly)	-2.26%	-3.12%	-3.59%	-3.83%	-3.83%	-3.83%
CVaR (1%, Monthly)	-3.46%	-4.51%	-5.39%	-6.58%	-7.49%	-8.16%
Hit Rate (Monthly)	60.81%	63.25%	62.16%	62.45%	62.19%	62.16%
Skewness	-0.03	0.04	-0.16	-0.41	-0.58	-0.66
Excess Kurtosis	7.98	8.39	10.79	15.89	21.91	26.63

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Intra-asset class combination: Equal risk versus equal weights

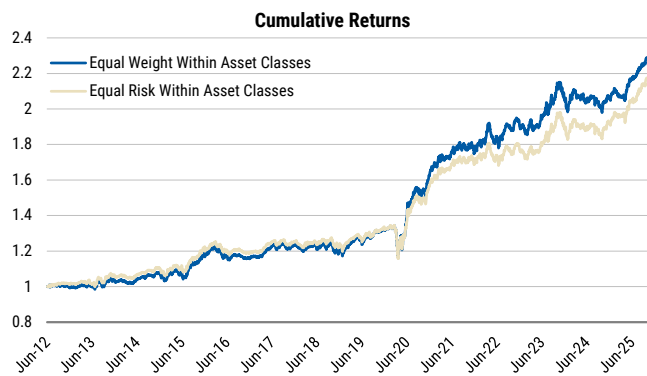
Next, we explore alternative approaches to portfolio construction. We begin by experimenting with different asset weighting schemes within each asset class prior to applying the realized skewness scaler. In our final implementation, we adopt an equal risk

weighting methodology, which aims to balance risk contributions across assets rather than capital allocations.

To estimate asset-level volatility, we use an exponentially weighted moving standard deviation with a long half-life of ten years. This choice reflects a deliberate design decision: shorter half-life parameters tend to overweight recent volatility spikes, often triggered by market distress, which can lead to underweighting assets precisely when skewness signals are most pronounced. Thus, using a shorter volatility estimation window would mean that volatility scaling at times works in the opposite direction than the skewness signal and possibly partially neutralizes it. However, using a long half-life smooths transient shocks and ensures that the portfolio remains responsive to realized skewness rather than reactive to short-term noise. At the same time, the inclusion of long-term volatility in portfolio construction allows us to balance the intra-asset class portfolio versus persistent volatility differences among assets.

We compare the equal risk weighting approach to a simple equal-weight scheme. [Exhibit 41](#) and [Exhibit 42](#) illustrate that both approaches deliver comparable risk-adjusted performance. We ultimately favor the equal risk framework, as there is no empirical evidence suggesting that assets with historically higher volatility consistently offer stronger skewness premia. By equalizing risk contributions, the strategy ensures more balanced attribution across assets and avoids overweighting those with elevated volatility that may not correspond to stronger signal quality.

Exhibit 41: Performance by intra-asset class weights



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

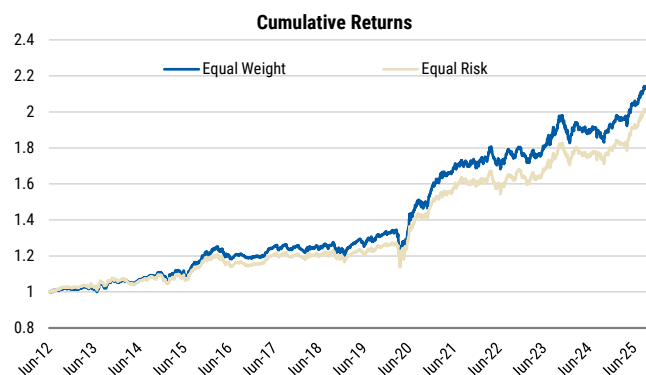
Exhibit 42: Performance statistics by intra-asset class weights

	Equal Weight Within Asset Classes	Equal Risk Within Asset Classes
CAGR	6.32%	5.91%
Annualized Vol	6.32%	5.88%
CAGR / Vol	1.00	1.01
Max DD	-13.66%	-13.95%
Max DD / Vol	-2.16	-2.37
Calmar Ratio	0.46	0.42
Sortino Ratio	1.44	1.45
VaR (1%, Monthly)	-4.13%	-3.83%
CVaR (1%, Monthly)	-6.63%	-6.58%
Hit Rate (Monthly)	61.44%	62.45%
Skewness	-0.41	-0.41
Excess Kurtosis	11.72	15.89

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Cross-asset class combination: Equal risk versus equal weights

At the cross-asset level, we also evaluate the impact of using equal weighting versus equal risk weighting across asset sleeves. The results, shown in [Exhibit 43](#) and [Exhibit 44](#) , indicate that both approaches yield similar performance on a risk-adjusted basis. Given that we already apply equal-risk allocation within each asset class, and that the asset sleeves exhibit comparable risk levels after applying the volatility scaler, we opt for the simpler and more transparent method of equal-weighting across sleeves.

Exhibit 43: Performance by cross-asset class weighting scheme

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 44: Performance statistics by cross-asset class weighting scheme

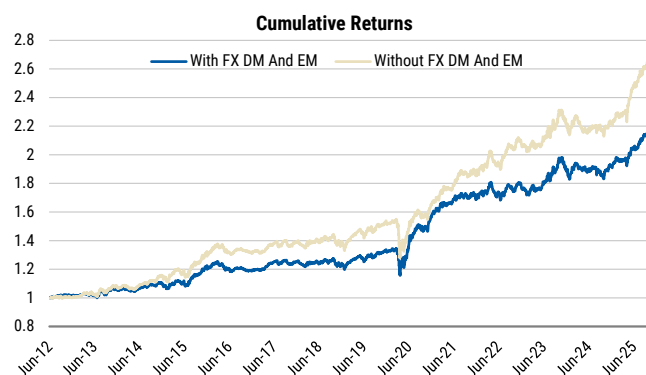
	Equal Weight	Equal Risk
CAGR	5.91%	5.44%
Annualized Vol	5.88%	5.41%
CAGR / Vol	1.01	1.01
Max DD	-13.95%	-10.89%
Max DD / Vol	-2.37	-2.01
Calmar Ratio	0.42	0.50
Sortino Ratio	1.45	1.46
VaR (1%, Monthly)	-3.83%	-3.41%
CVaR (1%, Monthly)	-6.58%	-5.20%
Hit Rate (Monthly)	62.45%	61.39%
Skewness	-0.41	-0.17
Excess Kurtosis	15.89	13.51

Source: Morgan Stanley Research, Bloomberg. Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Dropping FX sleeves

While we find modest positive CAGR/vol ratios in the FX sleeves of our strategy, the performance in FX markets falls behind the other asset classes. To assess their contribution, we evaluate the cross-asset portfolio's performance with and without the FX components. [Exhibit 45](#) and [Exhibit 46](#) compare these configurations.

The results show that excluding the FX sleeves leads to improvements in both cumulative returns and Sharpe ratios, suggesting that the FX allocation may dilute the overall signal quality. However, this comes at a cost that the portfolio experiences deeper drawdowns, and the drawdown-to-volatility ratio becomes more unfavorable. Notably, there are periods such as during Covid in 2020 where the inclusion of FX sleeves helped to cushion losses and contributed positively to overall portfolio resilience. While removing FX may enhance performance metrics in aggregate, retaining them can offer diversification benefits during periods of market stress.

Exhibit 45: Performance with and without FX

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 46: Performance statistics with and without FX

	With FX DM and EM	Without FX DM and EM
CAGR	5.91%	7.53%
Annualized Vol	5.88%	6.52%
CAGR / Vol	1.01	1.15
Max DD	-13.95%	-17.58%
Max DD / Vol	-2.37	-2.69
Calmar Ratio	0.42	0.43
Sortino Ratio	1.45	1.65
VaR (1%, Monthly)	-3.83%	-3.83%
CVaR (1%, Monthly)	-6.58%	-8.28%
Hit Rate (Monthly)	62.45%	62.82%
Skewness	-0.41	-0.68
Excess Kurtosis	15.89	17.43

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

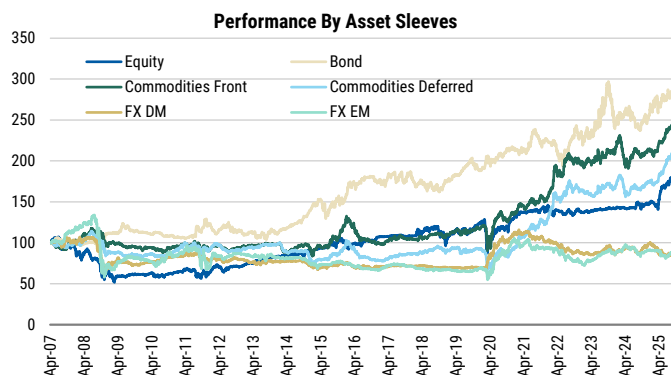
Extending backtest history

Our baseline backtest starts in 2012, which is a point in time when we have return data for most assets in our investment universe and burned in the estimates for the volatility scaling.

However, we are keen to also understand the long-term performance of our strategy using a reduced universe in the extended backtest. We are fully aware that the resulting strategy returns are only a proxy as they lack the investment depth of the full strategy universe.

In this extension exercise, we initiate the backtest in April 2007, aligning with the start date of the Trend Following strategy. Due to limited historical data for several underlying assets, the portfolio incorporates each asset only after it has accumulated at least six months of price history. [Exhibit 47](#) and [Exhibit 48](#) present the performance and statistical breakdown of each asset sleeve. As expected, some sleeves experienced significant drawdowns during the 2008 financial crisis, with the equity sleeve suffering the most pronounced impact. However, the bond sleeve is up by more than 10 percentage points in 2008. These observations underscore the value of diversification and the robustness of the strategy across different market regimes.

Exhibit 47: Extended backtest by asset class



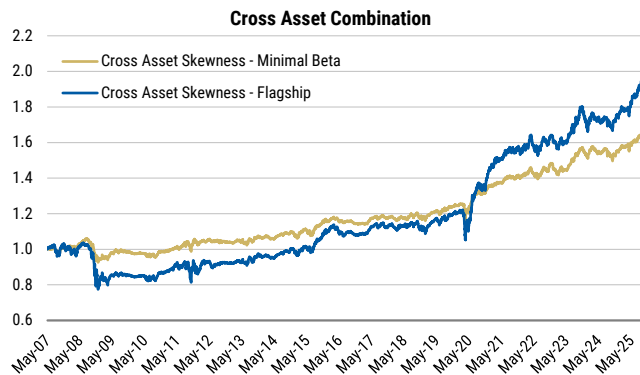
Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2007/04/16 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 48: Performance statistics for extended backtest by asset class

	Equity	Bond	Commodities Front	Commodities Deferred	FX DM	FX EM
CAGR	3.26%	5.83%	4.97%	4.07%	-0.85%	-0.60%
Annualized Vol	13.83%	10.32%	11.35%	10.96%	12.44%	15.61%
CAGR / Vol	0.24	0.56	0.44	0.37	-0.07	-0.04
Max DD	-51.74%	-20.20%	-31.59%	-41.57%	-40.91%	-59.10%
Max DD / Vol	-3.74	-1.96	-2.78	-3.79	-3.29	-3.79
Calmar Ratio	0.06	0.29	0.16	0.10	-0.02	-0.01
Sortino Ratio	0.33	0.81	0.63	0.51	-0.10	-0.05
VaR (1%, Monthly)	-12.86%	-6.94%	-9.02%	-10.10%	-9.32%	-15.02%
CVaR (1%, Monthly)	-18.18%	-8.35%	-12.02%	-13.94%	-15.39%	-24.42%
Hit Rate (Monthly)	61.47%	56.72%	55.22%	56.26%	50.43%	50.05%
Skewness	-0.27	0.05	-0.24	-0.63	-0.21	1.10
Excess Kurtosis	27.16	6.69	15.86	13.37	22.75	63.92

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2007/04/16 to 2025/09/30; all returns are net of all costs unless stated otherwise.

[Exhibit 49](#) and [Exhibit 50](#) display the performance at the cross-asset portfolio level. While the portfolio endured losses during the 2008 crisis period, it demonstrated resilience by recovering steadily and delivering strong performance in the subsequent years. We also find that the minimal beta implementation has a more resilient return in 2008 and a generally smoother return profile in line with its market-neutral objective.

Exhibit 49: Cross-asset extended backtest performance

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2007/05/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 50: Performance statistics for cross-asset extended backtest

	Cross Asset Skewness - Flagship	Cross Asset Skewness - Minimal Beta
CAGR	3.72%	2.79%
Annualized Vol	7.38%	4.00%
CAGR / Vol	0.50	0.70
Max DD	-25.22%	-12.68%
Max DD / Vol	-3.42	-3.17
Calmar Ratio	0.15	0.22
Sortino Ratio	0.70	0.99
VaR (1%, Monthly)	-6.70%	-2.87%
CVaR (1%, Monthly)	-9.83%	-4.81%
Hit Rate (Monthly)	59.24%	59.70%
Skewness	-0.43	-0.29
Excess Kurtosis	18.47	7.78

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2007/05/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

In this section, we explored a range of portfolio parameterizations and design choices to assess the robustness of the Realized Skewness strategy. Through systematic testing, we demonstrate that each configuration was carefully considered to balance performance, risk, and interpretability. The strategy consistently exhibits resilience across these design adjustments, with risk-adjusted returns remaining stable and well supported. These findings reinforce the robustness of the framework and its adaptability to different market environments and implementation constraints.

Conclusions

In this article we introduced our new cross-asset Skewness strategy. The signal in this strategy is realized skewness, which we calculate over several estimation and return windows to achieve a more robust signal. We obtain Sharpe ratios in excess of 1 by going short assets with a positive skewness and long assets with a negative skewness. We find that the scaling of the positions by the magnitude of the skewness scores provides a more intuitive signal. While the strategy follows a directional long/short approach, we find a rather limited overall beta to traditional assets. The approach works best in equities, rates, and commodities. Our results prove to be robust during a range of stability tests.

We run the strategy by asset class sleeve and constrain the respective market beta to ± 0.3 . We show that a tighter beta constraint of ± 0.05 still delivers Sharpe ratios around 1. Thus, the beta constraint parameter is a lever that we can pull in order to tilt the strategy characteristics between a more directional approach that blends in well into an asset allocation and a more market-neutral relative value return source. In both iterations our signal generates strong risk-adjusted returns.

In addition to being a strategy with strong performance in its own right, we find that the cross-asset Skewness strategy is a great diversifier to Trend Following. Correlations between the two strategies are low. Furthermore, we find that fundamentally the two diversify each other well too: Trend Following does well during prolonged drawdowns but tends to suffer when markets revert. The Skewness strategy, on the other hand, tends to go long assets as they reach the market trough when skewness tends to be negative but then benefits from the market rebound. A combination of Trend Following and Skewness can provide a more balanced performance during down markets and the subsequent recoveries.

We see Trend Following and Realized Skewness as two types of strategies that can serve as a core in a multi-asset long/short portfolio. They provide strong joint returns and diversify a portfolio of traditional assets, lifting up its risk-adjusted performance from 0.76 to 1.25.

Appendix

Overview of investment universe

[Exhibit 51](#) lists all 109 assets included in the strategy, organized by their respective asset classes.

Exhibit 51: Universe details

Equities	Bonds	Commodities Front	Commodities Deferred	FX DM	FX EM
FTSE 100	AUD 10y	Soybean Oil	Soybean Oil	AUD	BRL
ASX SPI 200	USD 10y	Corn	Cocoa	CAD	CLP
EURO STOXX 50	EUR 10y	Cocoa	Crude Oil	CHF	CNH
Swiss Market Index	JPY 10y	Crude Oil	Corn	DKK	CZK
Russell 2000	ITA 10y	Brent Crude Oil	Brent Oil	EUR	HUF
MSCI Singapore	CAD 10y	Cotton	Cotton	GBP	IDR
S&P Canada 60	GBP 10y	Feeder Cattle	Feeder Cattle	JPY	ILS
NASDAQ 100		Gold	Gold	NOK	INR
Nikkei 225		Copper	Copper	NZD	KRW
DAX		Heating Oil	ULS Diesel	SEK	MXN
S&P 500 Mid Cap		Coffee	Coffee Arabica		PHP
S&P 500		Kansas Wheat	Kansas Wheat		PLN
CAC 40		Aluminium	Aluminium		SGD
KOSPI 200		Live Cattle	Live Cattle		THB
FTSE MIB		Lean Hogs	Lean Hogs		TWD
Topix		Lead	Nickel		ZAR
Hang Seng		Nickel	Zinc		COP
FTSE/JSE Top 40		Zinc	Natural Gas		
Bovespa		Natural Gas	Sugar		
Dow Jones		Palladium	Low Sulphur Gas		
AEX		Gas Oil	Soybean Meal		
OMX Stockholm 30		Soybean	Soybean		
H-Shares Hang Seng China Enterprises		Sugar	Wheat		
FTSE China A50		Silver	RBOB Gasoline		
		Soybean Meal			
		Wheat			
		RBOB Gasoline			

Source: Morgan Stanley Research

Portfolio constituents weights distribution

We present in [Exhibit 52](#) the distribution of constituent weights within the flagship Realized Skewness portfolio. Each row for an asset class shows the corresponding sleeve's weight in the final cross-asset portfolio. Similarly, each row for an individual asset indicates its weight within the overall cross-asset portfolio.

Exhibit 52: Weights summary (equity and bond)

	Minimum	5th Percentile	Median	Mean	95th Percentile	Maximum
Equity	21.34%	24.34%	25.05%	25.04%	25.82%	26.97%
FTSE/JSE Top 40	-1.27%	-0.99%	0.19%	0.06%	1.12%	1.47%
Bovespa	-1.48%	-1.09%	-0.51%	-0.39%	0.61%	2.31%
CAC 40	-1.50%	-0.47%	0.59%	0.57%	1.72%	2.21%
Dow Jones (\$5) E-mini	-1.17%	-0.62%	0.98%	1.44%	4.41%	5.09%
AEX	-0.70%	-0.31%	0.69%	0.99%	2.62%	3.01%
S&P 500 E-Mini	-0.36%	0.19%	1.19%	1.30%	3.07%	3.53%
S&P 500 Mid Cap	-0.88%	-0.24%	0.73%	0.78%	2.24%	2.70%
OMX Stockholm 30	-0.90%	-0.28%	0.63%	0.70%	1.93%	2.47%
DAX	-0.95%	-0.25%	0.63%	0.60%	1.47%	2.05%
H-Shares Hang Seng China Enterprises	-2.97%	-1.55%	-0.06%	-0.25%	0.81%	1.95%
Hang Seng	-4.59%	-1.57%	0.54%	0.09%	1.53%	2.20%
KOSPI 200	-1.04%	-0.28%	0.56%	0.66%	2.02%	2.79%
Nikkei 225 mini JPX	-1.21%	-0.61%	0.79%	0.66%	1.91%	2.71%
NASDAQ 100 E-Mini	-0.38%	-0.10%	1.06%	1.02%	2.20%	2.70%
S&P Canada 60	-1.29%	0.16%	1.17%	1.24%	3.04%	3.72%
MSCI Singapore SGX	-1.25%	-0.64%	0.25%	0.49%	2.34%	3.45%
Russell 2000 eMini ICE/CME	-1.41%	-0.66%	0.51%	0.42%	1.32%	2.19%
Swiss Market	-0.60%	-0.06%	1.32%	1.36%	3.37%	3.87%
FTSE MIB	-1.14%	-0.75%	0.56%	0.50%	1.48%	2.43%
Topix	-1.12%	-0.41%	0.92%	0.88%	2.36%	3.08%
EURO STOXX 50	-1.24%	-0.56%	0.52%	0.53%	1.71%	2.07%
ASX SPI 200	-1.23%	-0.50%	1.21%	1.13%	2.81%	3.50%
FTSE China A50 SGX	-2.73%	-1.76%	-0.29%	-0.36%	0.72%	1.08%
FTSE 100	-1.58%	-0.82%	0.84%	0.81%	2.48%	3.06%
Bond	22.81%	24.08%	24.99%	24.99%	25.88%	28.70%
10yr Australian Gov. Bond	-8.95%	-4.02%	4.84%	5.17%	13.92%	18.67%
US 10-Year T-Note	-25.92%	-20.91%	0.17%	-0.40%	18.51%	24.95%
Euro-Bund 10-Year	-17.97%	-14.08%	7.29%	7.41%	29.11%	41.86%
Japanese 10-Year Bond JPX	-96.56%	-81.64%	33.61%	21.11%	70.98%	83.19%
Euro-BTP	-13.36%	-8.76%	7.35%	6.19%	19.75%	25.57%
10yr Canadian Gov. Bond	-22.62%	-13.78%	2.20%	2.35%	17.65%	25.96%
Long Gilt	-14.77%	-9.15%	2.57%	2.26%	13.31%	18.10%

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30.

Exhibit 53: Weights summary (commodities)

	Minimum	5th Percentile	Median	Mean	95th Percentile	Maximum
Commodities Front	10.80%	12.16%	12.50%	12.52%	12.95%	13.92%
Soybean Oil	-2.79%	-2.17%	-0.90%	-0.58%	1.70%	2.91%
Cocoa	-3.57%	-1.54%	-0.36%	-0.30%	1.13%	1.82%
Corn	-2.89%	-2.22%	-0.02%	-0.10%	1.39%	2.04%
Crude Oil	-1.43%	-1.07%	-0.01%	0.01%	1.08%	2.99%
Brent Crude Oil ICE	-1.86%	-1.22%	0.15%	0.21%	1.62%	3.14%
Cotton #2	-2.49%	-1.52%	0.08%	0.19%	2.16%	2.88%
Feeder Cattle	-1.59%	-1.08%	0.89%	0.88%	2.97%	4.10%
Gold COMEX	-3.66%	-2.85%	-0.14%	-0.22%	2.51%	4.25%
Copper High Grade COMEX	-2.65%	-1.67%	-0.22%	0.02%	2.79%	3.73%
Heating Oil #2 NYMEX	-2.18%	-1.59%	-0.18%	-0.04%	1.76%	3.15%
Coffee C	-3.37%	-2.30%	-1.01%	-1.16%	-0.27%	0.34%
Kansas Wheat	-2.81%	-2.14%	-1.12%	-1.15%	0.00%	0.87%
Aluminium LME	-4.81%	-3.09%	-1.24%	-0.91%	2.11%	3.68%
Live Cattle	-3.17%	-2.03%	0.73%	0.76%	3.43%	4.34%
Lean Hogs	-3.61%	-2.36%	-0.48%	-0.49%	1.19%	2.49%
Nickel LME	-2.49%	-2.04%	-0.53%	-0.48%	1.58%	2.72%
Zinc LME	-2.78%	-2.01%	-0.33%	-0.35%	0.79%	1.34%
Natural Gas NYMEX	-2.16%	-1.50%	-0.44%	-0.53%	0.19%	0.91%
Palladium NYMEX	-2.32%	-1.69%	-0.25%	-0.05%	1.91%	2.36%
Gas Oil NYMEX	-2.55%	-1.91%	0.16%	0.03%	1.81%	3.53%
Soybean	-3.17%	-2.38%	-0.50%	-0.50%	1.22%	1.82%
Sugar #11	-2.29%	-1.94%	-0.81%	-0.69%	1.28%	2.17%
Soybean Meal	-3.26%	-2.81%	-1.71%	-1.51%	0.48%	1.25%
Wheat CBOT	-3.04%	-2.34%	-1.26%	-1.27%	-0.31%	0.02%
RBOB Gasoline NYMEX	-1.15%	-0.70%	0.14%	0.32%	1.81%	2.78%
Lead LME	-1.92%	-1.19%	-0.23%	-0.21%	0.96%	2.22%
Silver COMEX	-2.06%	-1.28%	-0.33%	-0.14%	1.59%	3.19%
Commodities Deferred	10.84%	12.18%	12.51%	12.52%	12.93%	13.58%
Soybean Oil D6	-2.88%	-2.06%	-0.83%	-0.48%	1.92%	3.22%
Cocoa D6	-4.04%	-1.64%	-0.36%	-0.28%	1.30%	1.93%
WTI Crude Oil D6	-1.51%	-1.03%	0.74%	0.75%	2.13%	3.48%
Corn D6	-3.03%	-2.50%	-0.14%	-0.16%	1.70%	2.67%
Brent Oil D6	-1.63%	-1.05%	0.65%	0.71%	2.43%	3.70%
Cotton D6	-2.82%	-1.73%	0.81%	0.89%	3.08%	4.32%
Feeder Cattle Roll D5	-3.99%	-1.00%	1.20%	1.43%	3.98%	4.98%
Gold D6	-3.49%	-2.58%	-0.11%	-0.13%	2.41%	4.11%
High Grade Copper D6	-2.72%	-1.71%	-0.18%	0.07%	2.81%	3.72%
ULS Diesel D6	-1.92%	-1.39%	0.37%	0.42%	2.14%	3.97%
Coffee Arabica D6	-3.47%	-2.43%	-1.08%	-1.20%	-0.22%	0.44%
Kansas Wheat D6	-2.96%	-2.25%	-1.11%	-1.14%	0.06%	0.91%
Aluminium D6	-4.56%	-2.91%	-1.22%	-0.90%	2.11%	3.85%
Live Cattle Roll D5	-2.47%	-0.56%	2.38%	2.67%	6.01%	6.71%
Lean Hogs Roll D5	-4.22%	-1.98%	1.00%	0.91%	3.10%	4.52%
Nickel D6	-2.30%	-1.97%	-0.48%	-0.45%	1.58%	2.63%
Zinc D6	-2.16%	-1.89%	-0.15%	-0.18%	1.01%	1.72%
Natural Gas D6	-1.92%	-1.08%	0.22%	0.19%	1.27%	1.80%
Low Sulphur Gas	-2.24%	-1.45%	0.62%	0.53%	2.25%	4.30%
Sugar D6	-2.79%	-2.33%	-0.72%	-0.68%	1.72%	2.49%
Soybean Meal D6	-4.11%	-3.22%	-1.42%	-1.50%	0.60%	1.63%
Soybean D6	-2.90%	-2.19%	-0.06%	-0.12%	2.08%	3.27%
Wheat D6	-3.04%	-2.35%	-1.21%	-1.24%	-0.16%	0.25%
RBOB Gasoline D6	-1.46%	-1.15%	0.69%	0.67%	2.25%	3.39%

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30.

Exhibit 54: Weights summary (FX)

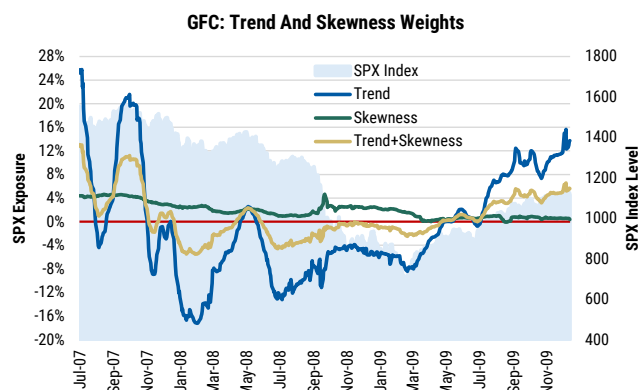
	Minimum	5th Percentile	Median	Mean	95th Percentile	Maximum
FX DM	11.26%	12.13%	12.46%	12.46%	12.77%	15.71%
EUR	-12.15%	-6.49%	-0.48%	-0.70%	3.61%	12.30%
GBP	-11.13%	-4.60%	2.76%	2.29%	12.96%	16.79%
JPY	-15.44%	-11.44%	-1.64%	-1.97%	9.47%	11.99%
CAD	-11.82%	-6.65%	0.28%	0.78%	10.44%	16.12%
CHF	-15.00%	-13.43%	-2.54%	-2.90%	5.72%	10.31%
AUD	-4.17%	-1.72%	0.59%	1.34%	7.33%	12.33%
NZD	-6.32%	-2.49%	0.63%	0.94%	4.91%	9.88%
NOK	-8.43%	-6.57%	0.12%	1.22%	9.59%	12.13%
SEK	-10.32%	-5.37%	0.33%	0.00%	5.64%	9.84%
DKK	-12.04%	-6.45%	-0.47%	-0.71%	3.62%	12.32%
FX EM	11.45%	12.12%	12.48%	12.47%	12.83%	13.47%
BRL	-4.22%	-3.02%	-0.05%	-0.11%	3.07%	5.15%
CLP	-6.92%	-3.31%	-0.32%	0.10%	5.28%	8.87%
CNH	-23.34%	-14.10%	6.88%	7.14%	35.94%	44.32%
CZK	-7.23%	-3.95%	0.75%	1.09%	6.65%	11.21%
HUF	-3.50%	-1.21%	0.71%	0.81%	3.47%	8.24%
IDR	-13.65%	-11.15%	-1.72%	-1.94%	8.10%	13.14%
ILS	-7.00%	-3.88%	1.37%	1.92%	12.16%	14.17%
INR	-17.30%	-4.69%	1.40%	1.44%	8.75%	14.08%
KRW	-13.25%	-6.76%	0.78%	0.26%	5.68%	8.23%
MXN	-2.27%	-0.40%	2.74%	3.13%	7.00%	9.27%
PHP	-4.13%	-1.77%	5.37%	5.62%	16.53%	19.66%
PLN	-2.49%	-1.05%	1.47%	1.54%	4.49%	9.07%
SGD	-13.94%	-9.83%	0.52%	0.90%	14.10%	17.77%
THB	-15.92%	-9.88%	-1.76%	-0.03%	11.64%	18.99%
ZAR	-2.90%	-1.27%	1.48%	1.43%	3.79%	4.74%
TWD	-24.69%	-19.87%	0.52%	-1.18%	14.00%	19.21%
COP	-5.12%	-2.23%	1.26%	1.44%	6.04%	9.85%

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30.

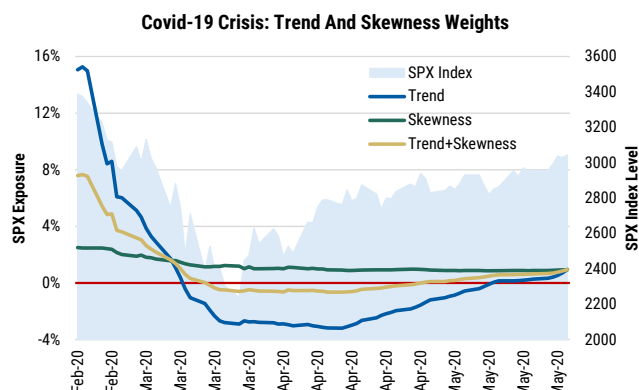
Skewness versus trend weights revolution during specific events

We extend the event study presented in [Performance during market distress](#) in [Interaction Between Skewness and Trend Following](#) by zooming in on S&P 500 and its corresponding weights within the Enhanced Trend Following and Realized Skewness portfolios during periods of market distress. Two case studies are highlighted below.

[Exhibit 55](#) illustrates the behavior of the Enhanced Trend portfolio during the global financial crisis. Initially, its S&P 500 exposures oscillated between positive and negative, but as losses deepened, the strategy maintained exposures near or below zero to stay defensive. When the rebound began, Trend promptly started adjusting weights, though it took time for exposures to turn positive. In contrast, Realized Skewness maintained a positive S&P 500 position throughout, helping the combined portfolio to remain broadly neutral before the recovery and then quickly leveraging up as the market exited distress. We observe similar dynamics during the Covid period in [Exhibit 56](#). Skewness reduced S&P 500 exposure during the drawdown but stayed positive to position for reversal, effectively offsetting Trend's negative stance and ensuring that the combined portfolio did not get hit during the rebound.

Exhibit 55: SPX exposures during the global financial crisis

Source: Bloomberg, Morgan Stanley Research

Exhibit 56: SPX exposures during the Covid crisis

Source: Bloomberg, Morgan Stanley Research

Add Skewness and Trend to a 60/40 portfolio – crises periods analysis

In [Exhibit 57](#), we compare the performance of a traditional 60/40 portfolio with two enhancements: adding Skewness alone and adding both Skewness and Trend. The results show that returns during crisis periods are significantly mitigated with these additions. Specifically, the average drawdown of 11% decreases to 7% when Skewness is included, and further improves to -5% with the addition of Trend.

Exhibit 57: Performance during crisis periods

Events	Start	End	60/40 Portfolio	Skewness + 60/40 Portfolio	Skewness + Trend + 60/40 Portfolio
GFC 2008 (DJIA Top - Lehman Collapse)	2007-07-19	2008-09-19	-13.31%	-7.49%	-5.91%
GFC 2008 (Lehman Collapse - Fed Fund Trough)	2008-09-19	2008-12-16	-16.67%	-13.89%	-2.41%
GFC 2008 (Fed Fund Trough - Global Stimulus)	2008-12-16	2009-04-02	-2.89%	-0.28%	-0.23%
GFC 2008 (Global Stimulus - Year End)	2009-04-02	2009-12-31	25.10%	9.71%	11.71%
GFC 2008	2007-07-19	2009-04-02	-29.85%	-20.57%	-8.38%
Flash Crash 2010	2010-05-05	2010-05-25	-5.09%	-3.07%	-3.35%
Greek Debt Crisis 2011	2011-07-22	2011-08-09	-7.33%	-4.50%	-4.02%
Taper Tantrum 2013	2013-05-22	2013-06-24	-5.50%	-3.28%	-4.90%
Chinese Stock Market Turbulence 2015	2015-08-17	2015-08-25	-5.67%	-1.95%	-2.45%
VIX Crash 2018	2018-01-26	2018-02-09	-5.73%	-2.75%	-4.65%
Q4 2018	2018-10-03	2018-12-24	-10.17%	-6.62%	-6.49%
COVID-19	2020-02-19	2020-03-24	-17.79%	-12.52%	-11.15%
January-June 2022	2022-01-01	2022-06-16	-17.96%	-9.47%	-0.96%
Trump Liberation Day	2025-04-02	2025-04-08	-7.26%	-3.60%	-4.51%
Crisis Returns Mean			-11.23%	-6.83%	-5.09%
Crisis Returns Median			-7.30%	-4.05%	-4.58%
1M After Crisis Mean			3.91%	2.71%	2.35%
1M After Crisis Median			3.07%	2.27%	2.17%
6M After Crisis Mean			10.99%	8.29%	7.80%
6M After Crisis Median			11.31%	7.46%	8.73%

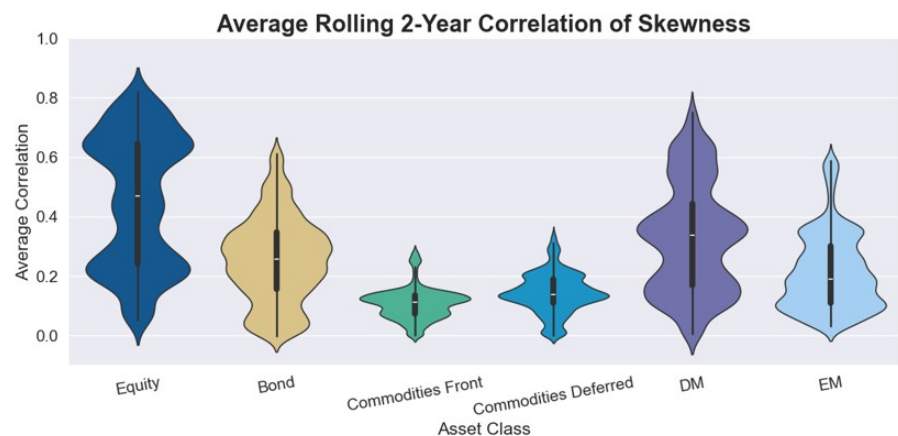
Source: Bloomberg, Morgan Stanley Research

Cross-sectional commodity Realized Skewness strategy

We add further analysis on the commodity Skewness strategy. First, we establish that the intra-asset class correlation of commodities is significantly smaller than in other asset classes. This greater dispersion in realized returns implies that skewness signals across individual commodities are less strongly correlated, reducing concentration risk and enhancing diversification potential within the sleeve. [Exhibit 58](#) illustrates this by showing the distribution of rolling correlations within each asset class, calculated over a two-year window of skewness signals. Equity and FX DM exhibit relatively higher signal correlations throughout the backtest period, indicating more synchronized behavior within these asset classes. In contrast, both front-month and deferred commodity contracts display very limited internal correlation, reflecting the fragmented nature of commodity markets.

The difference in correlation is important as it allows us to implement a cross-sectional commodity Skewness strategy. The reason is simple: in other asset classes the returns of the underlying assets as well as the skewness signals are highly correlated. When one asset is likely to go up, so are the others and vice versa. Thus, going long one asset and short the other essentially neutralizes the opportunity set. In commodities, however, the correlations across assets and signals are lower, allowing for a full cross-sectional implementation as usually signals and asset returns on the long and short side do not neutralize each other.

Exhibit 58: Distribution of intra-asset skewness correlations

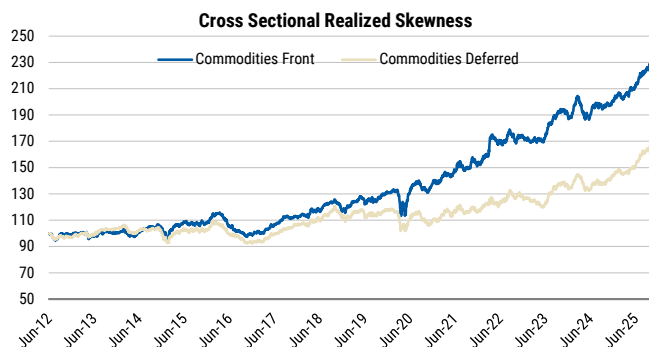


Source: Morgan Stanley Research

In [Exhibit 59](#) and [Exhibit 60](#), we present the performance of cross-sectional long/short strategies applied to both front-month and deferred commodity contracts. Each day, assets are ranked based on realized skewness, and positions are constructed by going long the low-ranked assets and short the high-ranked ones. Portfolio weights are determined using an equal risk weighting scheme, adjusted by the demeaned ranking scores. These weights are then normalized so that the sum of their absolute values equals one. Our findings indicate that this simple long/short approach delivers attractive risk-adjusted returns across both commodity classes. Notably, the strategy applied to front-month contracts exhibits superior performance, achieving a Sharpe ratio of 0.91 and maintaining a relatively contained maximum drawdown-to-volatility ratio of -2.28. These results

underscore the efficacy of skewness-based cross-sectional signals in generating consistent alpha within commodity markets.

Exhibit 59: Cross-sectional Skewness performance



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Exhibit 60: Performance statistics

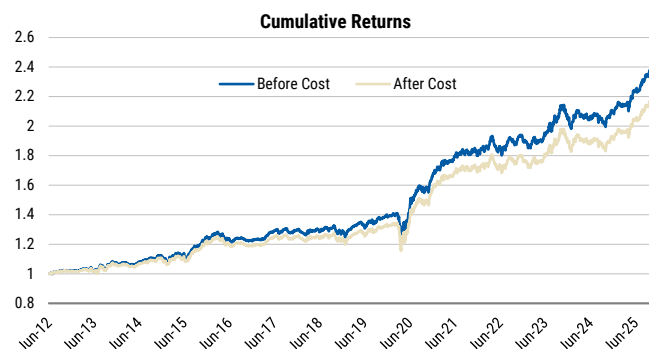
	Commodities Front	Commodities Deferred
CAGR	6.33%	3.81%
Annualized Vol	6.99%	6.70%
CAGR / Vol	0.91	0.57
Max DD	-15.94%	-16.04%
Max DD / Vol	-2.28	-2.39
Calmar Ratio	0.40	0.24
Sortino Ratio	1.32	0.80
VaR (1%, Monthly)	-4.75%	-5.58%
CVaR (1%, Monthly)	-7.58%	-7.52%
Hit Rate (Monthly)	60.93%	60.12%
Skewness	-0.04	-0.44
Excess Kurtosis	9.11	3.77

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30; all returns are net of all costs unless stated otherwise.

Gross of costs performance

We also report the flagship strategy's performance excluding transaction costs, alongside a comparison of pre- and post-cost results in [Exhibit 61](#) and [Exhibit 62](#). On an aggregate basis, annualized costs amount to approximately 73bp. This cost impact reduces the Sharpe ratio by 0.12, from 1.13 to 1.01.

Exhibit 61: Gross of costs performance



Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30.

Exhibit 62: Performance statistics

	Before Cost	After Cost
CAGR	6.64%	5.91%
Annualized Vol	5.88%	5.88%
CAGR / Vol	1.13	1.01
Max DD	-13.79%	-13.95%
Max DD / Vol	-2.34	-2.37
Calmar Ratio	0.48	0.42
Sortino Ratio	1.63	1.45
VaR (1%, Monthly)	-3.76%	-3.83%
CVaR (1%, Monthly)	-6.46%	-6.58%
Hit Rate (Monthly)	63.83%	62.45%
Skewness	-0.39	-0.41
Excess Kurtosis	15.83	15.89

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30.

[Exhibit 63](#) provides a detailed breakdown of cost impact by asset class sleeves. FX EM incurs the highest holding and daily trading costs, followed by front-month and deferred commodity contracts. The strategy's average two-way annual turnover is approximately 1,340%, since it is rebalanced on a daily basis.

Exhibit 63: Cost impact by asset class

		Equity	Bond	Commodities Front	Commodities Deferred	DM	EM	Equal Weight Combo
CAGR	Pre Cost	8.23%	7.31%	8.21%	7.17%	1.10%	2.27%	6.64%
	After Cost	7.81%	6.68%	7.42%	6.27%	0.79%	0.59%	5.91%
	COST (p.a.)	0.43%	0.63%	0.79%	0.91%	0.31%	1.67%	0.73%
CAGR/Vol	Pre Cost	0.79	0.67	0.73	0.66	0.11	0.22	1.13
	After Cost	0.75	0.61	0.66	0.58	0.08	0.06	1.01

Source: Bloomberg, Morgan Stanley Research; Note: Results reflect data from 2012/06/01 to 2025/09/30.

Strategies' Bloomberg tickers

Below, we provide the Bloomberg tickers for each variation and sub-sleeve of the strategy.

Exhibit 64: Strategies/sleeves' Bloomberg tickers

Strategy / Sleeves	Bloomberg Ticker
Morgan Stanley Cross Asset Skewness - Flagship	MSSKEWXA
Morgan Stanley Cross Asset Skewness - Flagship Excluding FX	MSSKEWXF
Equity	MSSKEWE0
Bond	MSSKEWB0
Commod Front	MSSKEWC0
Commodities Deferred	MSSKEWDC
FX DM	MSSKEWFD
FX EM	MSSKEWFE
Morgan Stanley Cross Asset Skewness - Minimal Beta	MSSKEWXB
Equity	MSSKEWEB
Bond	MSSKEWBB
Commod Front	MSSKEWCB
Commodities Deferred	MSSKEWDB
FX DM	MSSKEFDB
FX EM	MSSKEFEB

Source: Bloomberg, Morgan Stanley Research; The tickers listed in this table (MSSKEWXA, MSSKEWXF, MSSKEWE0, MSSKEWB0, MSSKEWC0, MSSKEWDC, MSSKEWFD, MSSKEWFE, MSSKEWXB, MSSKEWEB, MSSKEWBB, MSSKEWCB, MSSKEWDB, MSSKEFDB, MSSKEFEB) are Bloomberg tickers. The information contained herein has been prepared solely for informational purposes and is not a solicitation of any offer to buy or sell any security or other financial instrument or to participate in any trading strategy. Products and trades of this type may not be appropriate for every investor. Please consult with your legal and tax advisors before making any investment decision. Please contact your Morgan Stanley sales representative for more details.

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	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
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Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	567	15%	83	10%	15%	212	13%
Total	3,680		852			1610	

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